



Bachelor Thesis

Inpainting Detection and Localization via Vision Foundation Models

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Abstract

Image manipulations are becoming increasingly accessible and widespread due to the rapid advancement and diffusion of generative Artificial Intelligence, which allows for realistic and seamless modifications of digital images with minimal effort. Traditional forensic methods struggle to detect the manipulations produced by these modern AI tools, particularly those based on diffusion models. Malicious applications of these tools could facilitate the spread of disinformation and the falsification of digital evidence, posing a severe threat to information integrity. For these reasons, developing reliable methods to detect and localize image manipulations is becoming increasingly important.

This thesis explores a highly accessible methodology for inpainting localization by training a lightweight neural network on top of the output of DINOv3, a Vision Foundation Model utilized strictly as a frozen feature extractor. Multiple classification head architectures are investigated, including linear, MLP, and convolutional variants. These are evaluated across three backbone configurations, specifically ViT-S/16 alongside the Tiny and Small variants of ConvNeXt, as well as two inference strategies: a resize-based approach and a patch-based sliding-window technique that preserves full native resolution. Furthermore, ablation studies are conducted across objective functions and data augmentation pipelines to rigorously assess their impact on localization quality.

The models are trained and evaluated on a curated subset of the SAGI-D benchmark, containing images inpainted via five distinct modern generative pipelines. To assess generalization beyond the training distribution, two additional benchmarks are constructed, one derived from the public Beyond the Brush collection and one generated specifically for this thesis using an adapted version of the same automated pipeline. Crucially, both out-of-distribution benchmarks contain images generated by an advanced inpainting engine entirely absent from the training data.

The best configuration achieves strong in-distribution localization performance, reaching an F1 score of 0.854, an Intersection over Union of 0.745, and an AUROC of 0.970 on the held-out test split. While performance drops when evaluated against the out-of-distribution benchmarks, exposing the model to a multi-domain training set recovers a significant portion of this gap, more than doubling the F1 score on the previously unseen generative engine. These results indicate that the general-purpose representations of a frozen foundation model already contain substantial information about inpainting manipulations, enabling effective localization without training a large-scale architecture from scratch and making the proposed approach feasible on consumer hardware.

Keywords

Vision Foundation Model, Inpainting Detection, DinoV3, Digital Image Forensics, Inpainting generation, Artificial Intelligence

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Chapter 1

Introduction

In recent years, the rapid advancement of generative Artificial Intelligence has fundamentally transformed the landscape of digital media. Architectures such as Generative Adversarial Networks (GANs) and, more recently, Latent Diffusion Models have reached a level of sophistication where synthetic media is often visually indistinguishable from authentic photography. While these technologies have unlocked unprecedented creative possibilities, they have also democratized the ability to manipulate digital content. Advanced editing techniques, such as generative image inpainting, allow users to seamlessly add, remove, or replace specific elements within a photograph using simple text prompts, leaving the surrounding context perfectly intact.

This capability poses a severe threat to information integrity, providing a powerful vector for digital fraud, identity manipulation, and political disinformation. Consequently, the field of Digital Image Forensics is facing an urgent challenge: traditional detection tools, designed to spot manual cloning or splicing artifacts, are largely ineffective against the semantic coherence and high-frequency structural changes generated by modern AI. There is a critical need for advanced forensic models capable of not only detecting whether an image has been manipulated, but also precisely localizing the altered pixels.

Compounding this challenge is the sheer speed at which generative models themselves evolve. A forensic technique tuned to the artifacts of today’s state-of-the-art inpainting engine risks becoming obsolete within months, as newer, more sophisticated generators emerge and existing detectors need to be retrained, redesigned, or replaced entirely to keep pace. It is therefore not enough for a detection approach to perform well against current generative models: it must also be quick and cheap enough to adapt as the field moves forward, without requiring the kind of large-scale computational investment that only a handful of well-resourced laboratories can afford. This thesis is motivated precisely by this need for adaptability. Rather than pursuing the highest possible detection accuracy through increasingly large and specialized architectures, it investigates a promising and comparatively underexplored alternative: leveraging pretrained Vision Foundation Models as fixed, general-purpose feature extractors, so that adapting the resulting system to a newly released generative model requires retraining only a small, lightweight component rather than an entire specialized pipeline.

1.1 Image Inpainting: From Restoration Tool to Disinformation Vector

Image inpainting is the task of reconstructing missing, damaged, or deliberately removed regions of an image in a way that is visually and semantically consistent with the surrounding content. Borrowing its name from traditional art restoration, early computational approaches to inpainting were explicitly framed as restorative tools, designed to

fill small, localized defects rather than alter the semantic meaning of an image. Over the past two decades, the underlying technology has shifted from a restoration aid to a highly consequential generative capability. The introduction of deep generative models made it possible to seamlessly remove, replace, or insert entire objects while synthesizing a photorealistic context, and consumer software quickly incorporated these capabilities under names such as generative fill, effectively placing forensic-grade manipulation tools directly into the hands of casual users.

This shift presents a critical vector for disinformation. The ability to locally alter content while leaving the rest of the image untouched is particularly insidious, as it preserves the apparent authenticity of the original photograph. The deliberate removal or alteration of elements in authentic photographs has long been a highly sensitive issue in photojournalism, and a notable early example is the controversy surrounding renowned photographer Steve McCurry, who was found to have used manual cloning tools to seamlessly remove people and distracting objects from his published documentary photographs [9]. Such manual manipulation, however, inherently required significant technical skill, time, and professional software, and its consequences were largely confined to questions of editorial integrity. With the advent of generative AI, this technical barrier has entirely vanished, and the stakes have grown considerably higher: in January 2026, the White House posted on social media a photograph of a protester’s arrest in which her facial expression had been digitally altered to appear distressed and tearful, a change from the neutral expression visible in the original image, accompanying a caption that discredited her; a digital forensics expert consulted on the case noted that the incident was part of a broader, troubling pattern of AI-manipulated content that progressively erodes public trust in shared imagery [25]. Unlike the McCurry case, no particular skill or specialized software was required to produce this alteration, and it was published directly by an official government communication channel for explicitly political ends. Examples like these underscore the risks posed by the misuse of the latest inpainting technologies, highlighting how the democratization of state-of-the-art manipulation tools calls for an urgent need for detection tools capable of keeping pace with the latest generative models.

1.1.1 Challenges in Inpainting Forensics

Beyond the societal implications, the inpainting detection and localization task presents several structural challenges that heavily influenced the dataset and architectural choices of this thesis:

1. **Algorithmic Diversity:** Inpainting techniques span multiple architectures, from PDE-based inpainting and exemplar-based methods to modern large-scale latent diffusion models. Each family leaves distinct artifacts, meaning a model trained on one generation of methods may generalize poorly to another. Furthermore, the most capable modern systems are closed-source and accessible only through paid APIs, limiting the ability to inspect their internal parameters or systematically study their artifacts.
2. **Dataset Scarcity:** Unlike established computer vision tasks, inpainting forensics lacks a universally adopted benchmark. Furthermore, because of the rapid evolution of generative models, locally curated datasets risk becoming obsolete relative to the state of the art within a short timeframe.
3. **Absence of Ground-Truth Masks:** A considerable number of modern generative models perform full image-to-image regeneration rather than explicitly editing a well-defined local area. Since no binary mask of the modified region is produced, these advanced models cannot be utilized to build supervised localization datasets, restricting the available training data to methods that strictly preserve the unmodified background.
4. **Detection vs. Localization:** It is crucial to distinguish between *forgery detection* (a binary, image-level classification determining if an image is manipulated) and *forgery*

localization (a pixel-level task producing a dense map that identifies precisely which regions have been altered). Localization is a strictly harder and comparatively less-studied problem, and remains the primary objective of the framework proposed in this work.

1.2 The Internship and Collaborative Framework

This thesis was carried out within the Bachelor’s Degree Program in Artificial Intelligence, jointly offered by the University of Milan, the University of Milano-Bicocca, and the University of Pavia. The research and experimental work were conducted during a 225-hour internship in collaboration with IdentifAI Labs Spa, a company dedicated to the detection and recognition of digital manipulations and deepfakes, under the supervision of Daniele Bellavista.

The primary research focus of this project is the localization of generative inpainting. This thesis investigates whether a single, unmodified Vision Foundation Model inherently possesses enough latent knowledge to localize forgeries when paired with a minimal, lightweight classification head. While this approach might appear structurally simple, it offers substantial benefits in terms of adaptability and computational efficiency. A detailed overview of the motivations driving this methodology is provided in Section 1.3.

The same core problem was researched in parallel by my colleague Michelangelo Brusamarello, who explored a fundamentally different architectural path. His work develops a highly specialized, multi-branch neural network designed to explicitly extract and fuse distinct forensic signals, such as semantic context and low-level noise, into a comprehensive detection pipeline. Consequently, the two projects follow complementary architectural approaches. Rather than competing directly, they provide distinct insights into the same forensic challenge, contrasting the agility and accessibility of frozen foundation models with the deep feature extraction of customized multi-modal networks.

1.3 Scope and Contributions

The primary objective of this work is to identify an inpainting localization approach that performs well while remaining computationally accessible, rather than pursuing peak accuracy through an increasingly large and specialized architecture. Motivated by this goal, this thesis explores the potential of Vision Foundation Models, specifically the DINOv3 family of Vision Transformer and ConvNeXt backbones [42], described in detail in Chapter 4. In particular, we are interested in understanding whether the general-purpose representations learned by a foundation model, kept entirely frozen and never fine-tuned for the forensic task, already contain sufficient information to localize inpainting manipulations, and whether this holds even for comparatively small backbones such as DINOv3 ViT-S/16 (approximately 21 million parameters) and DINOv3 ConvNeXt-Small (50 million parameters). By leveraging these frozen models as feature extractors, we investigate how their rich, self-supervised spatial representations can be decoded by a minimal classification head to expose hidden forgery traces, without ever updating the backbone itself.

A central design goal of this research is hardware accessibility. Many recent forensic tools rely on massive, task-specific architectures that require industrial-grade GPU clusters for training and inference, placing them out of reach for independent researchers, journalists, or small organizations. The framework proposed in this thesis is explicitly engineered to be trainable and executable on consumer-grade hardware. All models, training runs, and high-resolution evaluations presented in this work were successfully performed on a single NVIDIA GeForce RTX 3050 GPU with 8 GB of VRAM. This strict computational constraint drove several key methodological choices, including the adoption of mixed-precision training, sliding-window inference, and the strict freezing of the foundation backbone.

Furthermore, this work includes a deep analysis of Out-of-Distribution (OOD) generalization, as well as ablation studies for the components introduced. Deep learning forensic models notoriously suffer from dataset bias, learning to recognize the specific noise of a training generator rather than universal forgery features. This thesis provides a comprehensive evaluation of how different backbone architectures and data augmentation strategies behave when exposed to unseen, highly sophisticated generative engines like Stable Diffusion XL.

1.4 Thesis Structure

Following this introduction to the core problem and the proposed methodology, the thesis is structured as follows:

- **Chapter 2** surveys the theoretical background and the state of the art. It traces the evolution of image inpainting techniques up to modern latent diffusion models. Subsequently, it examines how forensic techniques adapted to this generative evolution, shifting from classical statistical detectors to massive multi-modal networks. Finally, it introduces foundation-model-based paradigms as an emerging and highly accessible alternative, which directly motivates the approach proposed in this thesis.
- **Chapter 3** addresses the structural challenges of forensic data collection and details the datasets utilized for training and evaluation. It outlines the selection criteria for the SAGI-D training dataset and the diverse generative architectures it comprises. Furthermore, it introduces the automated Beyond the Brush pipeline, describing both the public Flickr30k dataset and our custom modifications used to generate an independent benchmark, explicitly clarifying how both are employed for out-of-distribution testing.
- **Chapter 4** formalizes the proposed DinoLizer methodology. It provides an architectural breakdown of the frozen DINOv3 Vision Transformer and ConvNeXt backbones, followed by the design of the lightweight classification heads. The chapter concludes with a comprehensive overview of the mixed-precision training pipeline and the respective inference strategies.
- **Chapter 5** presents the core experimental findings. It evaluates the in-distribution localization performance of the architectures, comparing resize-based and patch-based inference modes. The analysis is further supported by ablation studies on augmentation and objective functions and rigorous out-of-distribution tests to quantify cross-domain generalization.
- **Chapter 6** summarizes the key insights of the research. It discusses the limitations introduced by the strict computational constraints adopted throughout the project and proposes potential directions for future developments in generative manipulation forensics.

Chapter 2

Image Inpainting & Inpainting Detection

This chapter provides the theoretical foundation for the thesis by exploring the evolution of image inpainting and the corresponding forensic techniques developed to detect it. The remainder of this chapter is organized as follows. Sections 2.1 and 2.2 review the evolution of image inpainting techniques, from early diffusion and exemplar-based methods to the current generation of latent diffusion models. Finally, Section 2.3 surveys the parallel evolution of inpainting detection and localization methods, concluding with the recent foundation-model-based approaches that directly motivate the framework proposed in Chapter 4.

2.1 The evolution of Inpainting

The methods proposed for image inpainting over the past three decades can be grouped into six broad, roughly chronological families, summarized in Figure 2.1: methods rooted in partial differential equations (PDE-based), exemplar-based patch copying methods, convolutional autoencoders, Generative Adversarial Networks (GANs), Vision Transformer based methods, and, most recently, diffusion probabilistic models. Each new family did not simply replace the previous one, but rather addressed a specific limitation of its predecessors, progressively expanding the range and realism of achievable manipulations, and, as a direct consequence, the difficulty of detecting them.

As highlighted in Figure 2.1, these six families fall into two much broader categories: classical, hand-crafted methods, which rely on explicit mathematical models of color and texture, and deep-learning-based methods, which instead learn their priors directly from data. This division is necessary because deep learning has significantly improved the accuracy and effectiveness of image inpainting, representing a qualitative improvement over the classical methods. This also constitutes a turning point in the forgery-detection field, as previously top-performing detectors were ineffective against these new technologies.

Grouping inpainting methods based on their model structure can be very insightful, especially for tasks like inpainting detection, as different architectures present different strengths and limitations. Most importantly, different model families can exhibit characteristic artifact patterns. Knowing which family a given inpainting method belongs to is often a strong predictor of both its typical failure modes and, as discussed in Section 2.3, of how easily it can subsequently be detected.

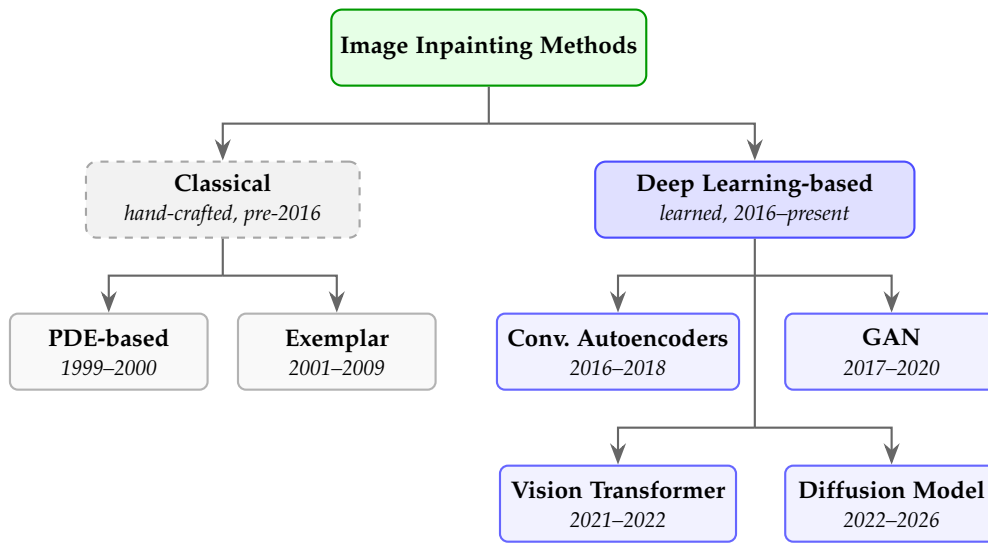


Figure 2.1: Evolution of the main families of image inpainting techniques discussed in this section, grouped into classical, hand-crafted approaches and deep learning-based approaches.

2.1.1 Classical, Hand-Crafted Methods

Partial Differential Equation (PDE) Based Methods

The earliest computational approaches to inpainting were rooted in partial differential equations and were directly inspired by the manual restoration of paintings. The core idea is to treat the missing region as a boundary-value problem: intensity and gradient information along the edge of the hole is propagated inward along the direction of the isophotes, the lines of constant image intensity, effectively continuing existing contours and structures into the gap [7]. The same propagation process was later reformulated as a diffusion problem governed by the Navier-Stokes equations that describe fluid dynamics, offering a more principled mathematical framework for controlling how information spreads inward while preserving sharp edges [6]. Because these methods only ever interpolate from the boundary, they work well for thin, elongated defects such as scratches, but they possess no notion of texture or repeated structure: as the size of the missing region increases, the propagated information simply blurs together, producing a smeared, out-of-focus patch rather than a plausible reconstruction.

Exemplar-Based (Patch-Based) Methods

A parallel line of work, exemplar-based or patch-based inpainting, addressed the texture problem directly by copying content from elsewhere in the same image rather than propagating it. Building on early non-parametric texture synthesis, where a missing pixel is filled by searching the rest of the image for the patch that best matches its known neighbourhood [16], later work formalized the process as a global optimization: the region is filled patch by patch, always prioritizing the patch whose surrounding structure is most reliable, typically located along strong edges, and for each one selecting the best-matching candidate found anywhere in the image [12]. Because an exhaustive search over every possible patch location is computationally expensive, later work reformulated the same idea as a randomized nearest-neighbour search that propagates good matches to neighbouring patches instead of restarting from scratch, making the search several orders of magnitude faster and enabling near-interactive use in mainstream editing tools [4]. Exemplar-based methods reproduce texture far more convincingly than PDE-based ones, since they reuse genuine image statistics rather than interpolating, but they have no notion of global se-

mantics: they can only recombine content that is already present in the image, and they tend to fail, or produce visibly repetitive and geometrically inconsistent patterns, whenever the missing region requires content, such as an object, that is not available anywhere else in the picture.

Both families described above share a common trait that sets them apart from every method discussed in the remainder of this section: they rely entirely on explicit, hand-crafted mathematical models of how color and texture should behave, with no component learned from data. Every technique discussed from this point onward, from convolutional and adversarial networks to Vision Transformers and diffusion models, instead belongs to the deep learning paradigm, and learns its priors directly from large collections of images rather than from an explicit mathematical formulation; this shift, and its consequences for the forensic community, is discussed further in Section 2.3.

2.2 Deep-Learning Based Image Inpainting

Rather than relying on explicit hand-crafted mathematical models of how color and texture should behave, deep-learning based approaches learn their priors directly from large collections of images. This qualitative shift made modern inpainting significantly more capable than its classical predecessors, transforming it from a localized restoration technique into a considerably harder to detect semantic generation tool. The remainder of this section reviews this evolution chronologically, demonstrating how each architectural breakthrough was designed to explicitly solve the structural limitations of the previous generation, culminating in the highly realistic diffusion models that dominate the current landscape.

2.2.1 Convolutional Autoencoders

One of the first influential deep-learning approaches applied specifically to inpainting was the convolutional autoencoder architecture, introduced for this task by Context Encoders [39]. This network consists of two main components: an encoder and a decoder, both built entirely from convolutional layers. The encoder processes the masked input image through a series of convolutions and pooling operations, progressively reducing the spatial dimensions of the image while increasing the depth of the feature channels. This forces the network to compress the visual information into a dense, lower-dimensional latent representation, effectively capturing the global semantic context of the scene. The decoder then takes this compressed representation and applies transposed convolutions (or upsampling layers) to progressively rebuild the spatial dimensions, ultimately producing a reconstruction of the image, $\mathbf{x}_r$, including the predicted content within the missing region [39].

Early convolutional autoencoders were optimized using a standard pixel-wise reconstruction loss, such as the Mean Squared Error (L_2 distance) or Mean Absolute Error (L_1 distance) between the predicted pixels $G(\mathbf{x}_m)$ and the ground truth $\mathbf{x}$. Crucially, to prevent the network from wasting capacity reconstructing pixels that are already known, this penalty is strictly confined to the missing region. This is achieved by introducing a binary mask M (where 1 indicates the missing hole and 0 the authentic background) and computing the Hadamard product ($\odot$) between the mask and the pixel-wise difference:

$$\mathcal{L}_{\text{rec}} = \|M \odot (\mathbf{x} - \mathbf{x}_r)\|^2. \quad (2.1)$$

While mathematically straightforward, this objective function presents a severe structural limitation widely documented in early literature [39, 22]. When a missing region is large, there are multiple semantically plausible ways to fill the texture. Because the network is penalized for deviations from one specific ground truth, it learns to output the mathematical average of all possible plausible textures to safely minimize the expected error.

Consequently, autoencoders trained solely with a reconstruction loss produce severely blurry and washed-out regions that lack high-frequency details, rendering the manipulation visually obvious and practically useless for generating realistic textures [39].

2.2.2 The Adversarial Paradigm: GAN-Based Inpainting

To overcome the blurriness inherent in pixel-wise reconstruction, researchers introduced the Generative Adversarial Network (GAN) training strategy to the inpainting task [39, 18, 3]. Instead of relying solely on the reconstruction loss, a second neural network, known as the Discriminator (D), was introduced and trained concurrently with the Generator (G).

The training process is formulated as a minimax game where the generator and the discriminator are optimized jointly according to the following objective:

$$\min_G \max_D V(G, D) = \mathbb{E}_x [\log D(x)] + \mathbb{E}_{x_m} [\log(1 - D(G(x_m)))]. \quad (2.2)$$

In this formulation $\mathbb{E}_x$ represents the expected value over the distribution of authentic images and $\mathbb{E}_{x_m}$ represents the expected value over the masked inputs. The term $D(x)$ is the probability estimated by the discriminator that a real image is authentic, while $D(G(x_m))$ is the probability that the synthesized reconstruction is real. The discriminator is trained to maximize the objective function $V(G, D)$ to become better at detecting forgeries, while the generator is trained to minimize the same objective to fool the discriminator. The structural flow of this adversarial dynamic is visually summarized in Figure 2.2. By adding this adversarial penalty to the training pipeline, the generator is forced to commit to one specific and sharp texture rather than a blurry average. Global semantic consistency was later improved by pairing a local discriminator with a second image-level discriminator that evaluates the picture as a whole, discouraging solutions that look convincing up close but are inconsistent with the broader scene [22].

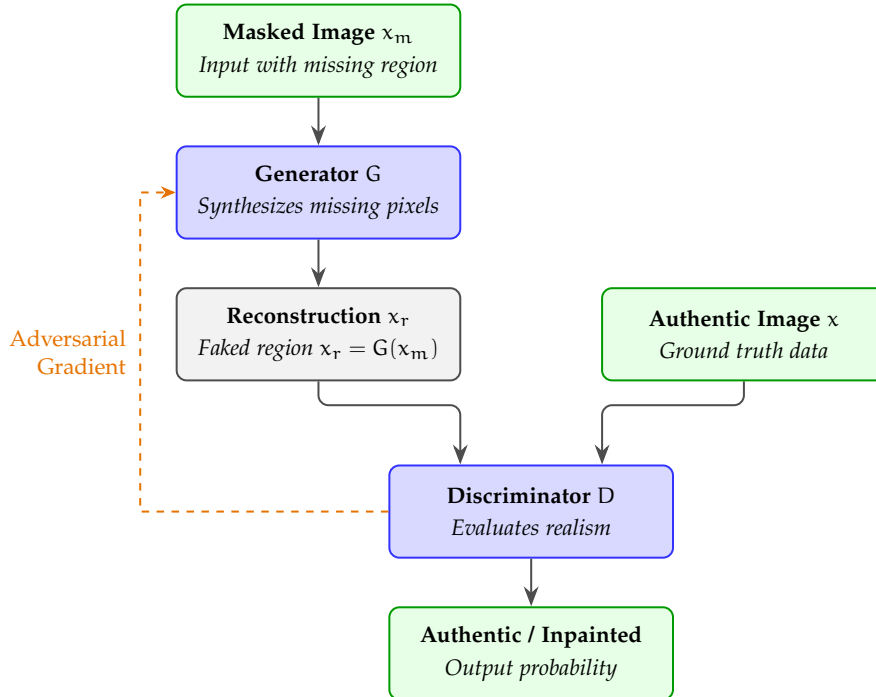


Figure 2.2: Conceptual representation of the adversarial training framework for CNN-based inpainting. The generator G and discriminator D compete, forcing the network to produce sharp textures.

Despite producing semantically plausible structures, this generation of CNNs suffered from a structural flaw caused by the convolution operation itself. Standard convolutional filters treat every pixel in their receptive field identically, so during the forward pass, meaningless values from the masked hole get mixed with valid background pixels, causing color bleeding and visual artifacts at the inpainting boundaries. Partial Convolutions were introduced to solve exactly this problem [29]: each filter computes its output using only the valid pixels present in its receptive field, rescaling the result accordingly, while a binary mask tracking which locations are valid is updated and passed to the next layer, so that the corrupting influence of the hole is explicitly reduced as information propagates through the network.

The same research group then tackled a second, complementary problem: even once the hole's corrupting influence is suppressed, the network still has no explicit mechanism to actively borrow useful content from elsewhere in the image, relying only on whatever context survives the encoder's bottleneck. Contextual Attention addresses this by letting the network directly search for and copy relevant background content [50]: for every patch inside the hole, a similarity score is computed against every patch outside it, and these scores are used to blend in the most relevant background texture before it is refined by further convolutional layers, giving the network an explicit, copy-based shortcut on top of what it learns to generate on its own.

Partial Convolutions, however, update their validity mask using a fixed, hand-designed geometric rule, which cannot adapt to what the network actually finds useful at each layer. Gated Convolutions generalize this idea by replacing the fixed rule with a learned, per-channel soft gate [51]: instead of a binary valid/invalid mask, the network itself learns, at every layer and every spatial location, how much of a given feature should be allowed to propagate forward, letting it dynamically prioritize the most informative signals rather than following a rigid geometric criterion.

A final, structurally different strategy tackles the problem at a higher level of abstraction rather than refining the convolution operation itself: instead of generating color and texture directly, EdgeConnect first predicts a plausible set of structural edges spanning the missing region, and only afterwards generates the corresponding texture conditioned on those edges [37], decoupling the comparatively hard problem of inferring global structure from the easier problem of filling in local appearance once that structure is known.

2.2.3 Vision Transformers and Global Context

Despite the sophisticated architectural patches applied to convolutional networks, their inherently localized receptive field remained a fundamental bottleneck for reconstructing large and complex masks. A true paradigm shift was required to effectively capture long-range dependencies across an image. This breakthrough was catalyzed by the unprecedented success of Transformer architectures in the field of Natural Language Processing. By conceptualizing an image not as a spatial grid of pixels but as a sequence of visual tokens, researchers successfully adapted this framework to computer vision, introducing the Vision Transformer (ViT) [15].

The core mathematical engine of this architecture is the scaled dot-product attention mechanism. The input image is divided into a sequence of non-overlapping patches. These patches are flattened and linearly projected into three distinct representations: Queries (Q), Keys (K), and Values (V). The self-attention operation is then defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad (2.3)$$

where d_k is the scaling dimension of the key vectors. The matrix multiplication QK^T calculates a similarity score between every single patch and every other patch in the image simultaneously. This creates a dense attention matrix that explicitly dictates how much visual context (V) each token should aggregate from the rest of the scene. This global

reasoning capability completely bypassed the spatial limitations of CNNs, making transformers exceptionally well suited for resolving large-scale structural ambiguities.

However, repurposing this architecture for image inpainting introduces a critical technical challenge. When an image is heavily masked, the tokens corresponding to the missing regions contain no valid visual information. Standard self-attention would blindly integrate these empty tokens into its similarity calculations, inevitably corrupting the attention matrix and degrading the global feature representation. To resolve this, specialized architectures such as the Mask-Aware Transformer (MAT) explicitly modified the standard attention formulation [26]. By introducing a mask-aware attention mechanism that prevents invalid or masked regions from contributing to the attention computation during the encoding phase, the network ensures that the generation process is guided strictly by the authentic background.

The ultimate objective of these architectures is to accurately predict the pixel values of the missing regions. To achieve this, the network outputs a full-resolution prediction which is optimized using a joint loss function. Instead of relying on a simple reconstruction penalty, models typically combine a masked L_1 distance loss to guarantee overall spatial fidelity with an adversarial loss to enforce the generation of sharp and realistic high-frequency textures. Formally, the joint objective is defined as:

$$\mathcal{L}_{\text{joint}} = \lambda_{L1} \|M \odot (x - x_r)\|_1 + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}(x_r), \quad (2.4)$$

where λ_{L1} and λ_{adv} are hyperparameters balancing the two terms. This mathematical combination ensures that the global structural coherence provided by the self-attention mechanism is successfully translated into photorealistic pixel values.

While Vision Transformers drastically improved structural coherence on large holes, they introduced a severe computational bottleneck. Because the attention matrix requires computing the similarity between every possible pair of N tokens, the computational and memory complexity scales quadratically as $\mathcal{O}(N^2)$. Consequently, doubling both spatial dimensions increases the number of tokens by a factor of four and the memory required for quadratic attention by a factor of sixteen. This quadratic scaling makes high-resolution pixel-space inpainting with pure transformers computationally prohibitive [3].

2.2.4 Diffusion Probabilistic Models

Current state-of-the-art solutions for generative image manipulation are overwhelmingly dominated by Diffusion Probabilistic Models (DPMs). Unlike previous architectures that attempt to reconstruct the missing pixels in a single forward pass, diffusion models frame generation as the process of reversing a gradual, thermodynamic-inspired noising process [21].

The rationale is built upon two opposing Markov chains. The forward process gradually destroys the structure of a pristine image x_0 by injecting Gaussian noise over T discrete timesteps. The amount of noise added at each step is controlled by a variance schedule β_t , acting as a hyperparameter. Each transition is mathematically defined as:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}). \quad (2.5)$$

Because the noise added at each individual step is itself normally distributed, it is possible to sample the noisy image directly at any arbitrary timestep t without simulating every intermediate step; for a sufficiently large T , the final state x_T becomes indistinguishable from pure Gaussian noise.

The generative capability lies in the learned reverse process. A neural network is trained to approximate the reverse transitions $p_\theta(x_{t-1} | x_t)$, effectively learning to denoise the image. Rather than predicting the clean image directly, the most successful formulation tasks the network with predicting the specific noise vector ϵ that was added to the image at timestep t [21]. The training objective is therefore framed as a supervised

regression problem, optimized using a simple Mean Squared Error (MSE) loss between the actual injected noise ϵ and the network's prediction ϵ_θ :

$$\mathcal{L}_{\text{diffusion}} = \mathbb{E}_{x_0, \epsilon, t} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]. \quad (2.6)$$

Architecturally, the noise predictor ϵ_θ is almost universally implemented as a U-Net. This network utilizes a ResNet-based encoder to compress the spatial resolution and extract deep features, followed by a symmetric upsampling decoder. Skip connections link the encoder and decoder at each spatial resolution, allowing the network to preserve fine, high-frequency details that would otherwise be lost during compression.

A defining characteristic of modern diffusion models is their ability to condition the generation process on textual descriptions (prompts) [41]. The user's text is processed by a pretrained text encoder, such as CLIP-based encoders in several Stable Diffusion models, which converts the prompt into a sequence of dense vector embeddings. These embeddings are injected into the U-Net through cross-attention layers. In this attention mechanism, the spatial feature maps of the image act as *queries*, while the text embeddings act as *keys* and *values*. This allows different spatial regions of the image to selectively "attend" to specific words in the prompt, enabling the generation of highly complex and semantically accurate content.

Latent Diffusion and Inpainting Logic

Operating directly in pixel space requires the U-Net to process massive spatial maps hundreds of times during a single generation, a computationally prohibitive task for high-resolution images. To solve this bottleneck, Stable Diffusion introduced the Latent Diffusion Model (LDM) [41]. The pipeline leverages a pre-trained Variational Autoencoder (VAE) to compress the pristine source image into a highly efficient, lower-dimensional latent representation. This encoding preserves the essential semantic information required for reconstruction while substantially reducing the spatial dimensionality, shifting the generative workload into a smaller mathematical space and cutting significantly the computational costs.

Applying a pretrained latent diffusion model to inpainting does not necessarily require training a dedicated model from scratch. RePaint [33] showed that an unconditional diffusion model can be repurposed for the task simply by resampling the known region of the image, using its true pixel values re-noised to the current timestep, at every reverse step; Figure 2.3 illustrates this specific resampling strategy.

Once the data is mapped into this efficient latent space, the generative inference process begins. Rather than modifying the source image directly, generation starts from a completely random tensor of pure Gaussian noise, denoted as the Initial Noise z_T . The core engine of the architecture is the U-Net Predictor, which performs an iterative denoising process. Over a predefined sequence of T reverse steps, the U-Net analyzes the current noisy latent and predicts the specific noise distribution that must be subtracted to recover a slightly cleaner version of the latent for the next step ($t-1$). To ensure the synthesized content aligns with the user's intent, a Text Prompt is processed by a frozen CLIP Text Encoder; these extracted embeddings are continuously injected into the U-Net as a conditioning signal to guide the semantic trajectory of the denoising.

If the U-Net were allowed to denoise the entire z_T tensor freely, it would generate a completely new scene from scratch, ignoring the original photograph entirely. To constrain the generation strictly to the missing areas, image inpainting employs a localized feedback mechanism at the end of every single step, known as Masked Blending [33]. While the U-Net predicts the denoised latent for the masked region, the authentic background must be mathematically matched to that exact same noise level. To achieve this, the original compressed latent z_0 (obtained from the VAE) is intentionally injected with forward Gaussian noise corresponding to the exact current timestep $t-1$, yielding the Known-Region Latent.

Using the binary mask M as a spatial stencil, the Masked Blending block physically stitches these two parallel realities together. The latent values inside the target region are

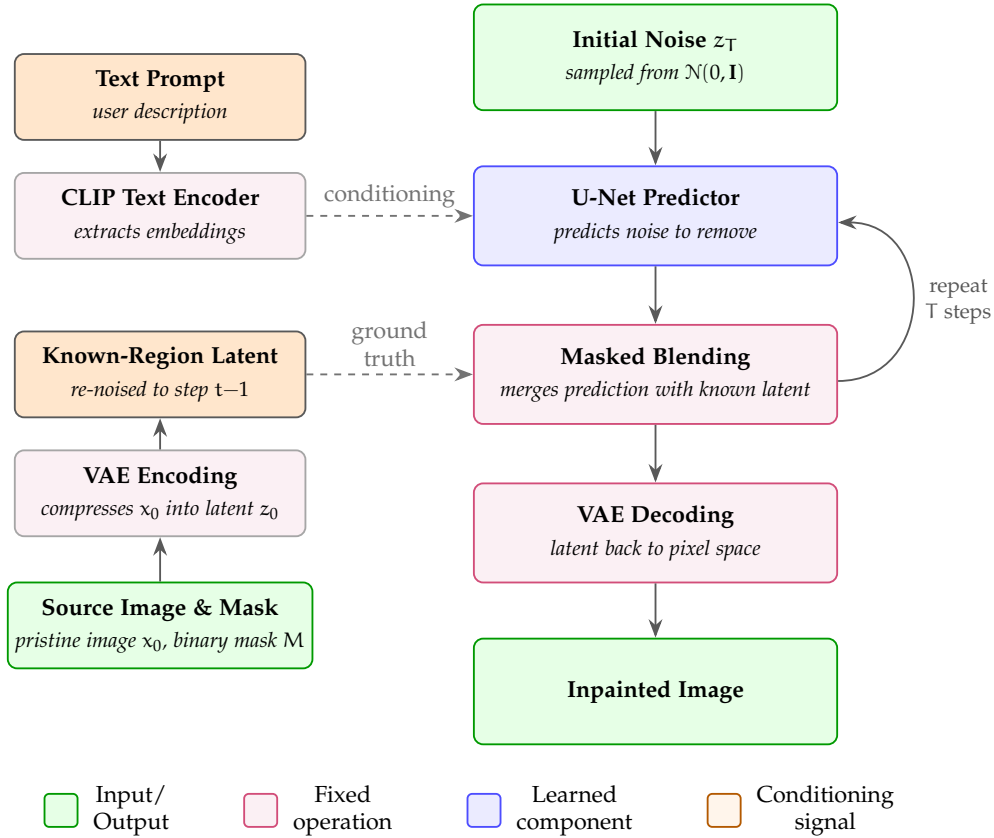


Figure 2.3: RePaint-based latent inpainting pipeline [33]. The model starts by iteratively denoising pure Gaussian noise, conditioned by a textual prompt. At each step, the U-Net’s prediction is merged via a binary mask with the authentic source image, which has been compressed by the VAE and explicitly re-noised to the matching timestep. After T steps, the fully blended latent is decoded back to pixel space to obtain the final inpainted result.

populated by the U-Net’s denoised prediction, while the values outside the region are strictly overwritten by the ground-truth Known-Region Latent. This composite, forcefully harmonized latent is then looped back into the U-Net for the subsequent denoising step.

By repeating this strict localized blending for all T steps, the newly hallucinated content is iteratively forced to align perfectly with the color, lighting, and semantic boundaries of the true background. Only when the loop concludes at step 0, meaning the noise has been progressively and completely removed, is the final composite latent passed through the VAE Decoding block. This decoder projects the representation back into the high-resolution pixel space to output the final Inpainted Image.

To ensure robust generalization and avoid overfitting to the specific artifact distribution of a single architecture, the experimental datasets curated for this work incorporate manipulations generated by multiple modern diffusion models. A detailed technical overview of the selected generators is provided in Chapter 3 (Section 3.6). These architectures generate new content within the masked region rather than simply copying existing pixels from the surrounding image, limiting significantly the effectiveness of traditional forensic detection methods, which prevalently rely on identifying blurred edges or checkerboard artifacts. They leave behind only the subtle, high-frequency systemic traces of the latent decoding process itself. This evolution fundamentally dictates the necessity for robust, open, and reproducible detection frameworks. This requirement serves as the primary motivation behind the foundation-model-based approach proposed in this work.

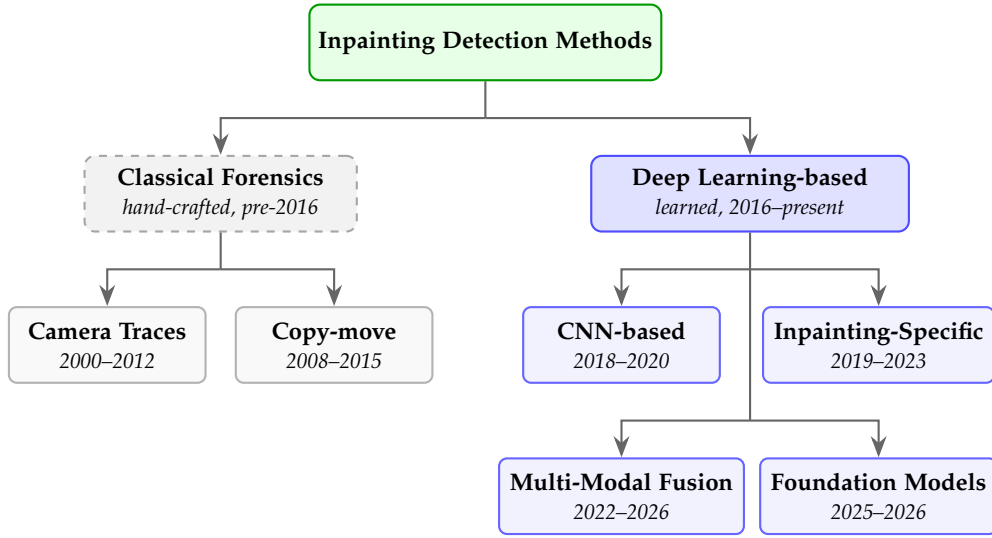


Figure 2.4: Evolution of the main families of inpainting detection and localization techniques, grouped into classical forensic approaches and deep learning-based approaches.

2.3 Evolution of Inpainting Detection and Localization

The transition from classical, hand-crafted inpainting techniques to deep learning-based methods, discussed in Sections 2.1 and 2.2, represented a critical turning point not only for image inpainting itself, but for the entire field of inpainting forensics. Classical detectors were built around the specific, mathematically predictable traces left by diffusion-based and exemplar-based inpainting, such as blurred boundaries or duplicated patches; once inpainting itself became a learned, data-driven process capable of synthesizing statistically and semantically coherent content, these assumptions largely stopped holding, and an entirely new generation of learned detectors became necessary to keep pace. The forensic techniques developed in response evolved through roughly the same number of stages as the generative methods that motivated them, summarized in Figure 2.4: classical statistical forensics, general-purpose CNN-based manipulation detectors, inpainting-specific localization networks, multi-modal networks, and, most recently, approaches built on Vision Foundation Models.

2.3.1 Classical Forensic Approaches

Long before deep learning became the dominant paradigm, image forensics relied on detecting subtle, low-level statistical inconsistencies left behind by the image formation and processing pipeline, rather than on recognizing manipulated content directly. Every digital camera introduces a distinctive noise pattern during acquisition, caused by minute imperfections in the silicon wafer of the sensor. This pattern, known as Photo Response Non-Uniformity (PRNU), is commonly used as a device-specific fingerprint. The sensor output I can be mathematically approximated as:

$$I = I_0 + I_0 K + \Theta, \quad (2.7)$$

where I_0 is the ideal noise-free image, K is the PRNU fingerprint matrix, and Θ represents independent random noise components. Because K is consistent across an authentic photograph, any localized disruption in its estimation reliably indicates that new pixels have been pasted or synthesized into the scene [34].

A closely related family of methods targets the recompression artifacts introduced by lossy formats such as JPEG. When a manipulated region is copied from a differently

compressed source, or when the whole image is saved a second time after editing, the periodic patterns left by the 8×8 Discrete Cosine Transform (DCT) blocks become misaligned or duplicated. By applying frequency-domain analysis, forensic algorithms can detect grid discontinuities that highlight the tampered bounding box. Object removal via early exemplar-based inpainting was historically addressed through copy-move detection algorithms. Since these generative methods worked by aggressively cloning patches from the known background, forensic tools utilized keypoint matching algorithms, such as SIFT [2], or dense block-matching to search for pairs of regions with statistically impossible levels of similarity, flagging them as candidates for a copy-paste operation.

These classical techniques are computationally inexpensive and require no training data. However, they are tailored to very specific traces and are generally less effective when the manipulated content is synthesized rather than copied from another image, because the characteristic correspondences exploited by copy-move methods are no longer necessarily present. Latent diffusion models synthesize entirely new content from Gaussian noise rather than copying existing pixels, and they are explicitly optimized to match the global statistical distribution of the image, significantly limiting the effectiveness of PRNU and block-matching detection strategies [3].

2.3.2 General-Purpose CNN-Based Detectors

The limitations of classical statistics drove the field toward deep learning, introducing general-purpose manipulation detectors capable of learning complex forensic features directly from data. Rather than relying on a single handcrafted test, Convolutional Neural Networks (CNNs) were trained to identify the boundary between authentic and manipulated pixels.

One highly influential architecture, ManTra-Net [49], frames the problem as local anomaly detection. The network extracts a dense feature representation of the image using a VGG-based backbone. Instead of predicting a binary class directly, it calculates a local anomaly score by evaluating how much the features of a specific patch deviate from the dominant feature distribution of the rest of the image. This self-referential strategy was designed to improve generalization across different manipulation types.

A complementary strategy focuses on extracting and evaluating camera-level noise through learned representations. The Noiseprint architecture [11] abandons standard classification in favor of a Siamese network approach. The network is trained using a contrastive loss function that minimizes the distance between image patches originating from the same camera model while maximizing the distance between patches from different cameras. This forces the CNN to suppress the semantic scene content completely and output a representation which emphasize camera-related noise residuals. Tampering is subsequently flagged wherever this extracted Noiseprint becomes locally inconsistent.

While significantly more robust than their classical predecessors, these general-purpose CNNs often struggle with state-of-the-art generative inpainting. Because they are optimized to recognize manipulations in general, they lack the specific inductive biases required to detect the extremely subtle, high-frequency systemic traces left by latent diffusion decoders. This motivates the need for architectures specialized for inpainting detection.

2.3.3 Inpainting-Specific Localization Networks

To close this specificity gap, a distinct line of research treats inpainting localization as an independent, highly specialized pixel-level segmentation problem. These architectures deliberately suppress the visual content of the image at the very first layer, forcing the network to analyze raw signal residuals rather than textures or shapes.

A rigorous example of this methodology is IID-Net [48]. This architecture structures the detection pipeline into three specialized stages: an enhancement block, an extraction block, and a decision block.

The enhancement block is designed to deliberately suppress the semantic content of the image, which standard convolutional filters naturally prioritize, to expose the microscopic traces left by generative inpainting. To achieve this, it fuses three distinct filtering strategies. First, a steganalysis-inspired Spatial Rich Model (SRM) filter captures localized noise distributions. Second, a fixed high-pass pre-filtering layer isolates high-frequency residuals. Finally, an adaptive Bayar convolutional layer learns low-level prediction residuals dynamically. Unlike standard convolutions, the Bayar layer explicitly constrains its kernel weights during the forward pass:

$$\sum_{(i,j) \neq (0,0)} w_{i,j} = 1, \quad w_{0,0} = -1. \quad (2.8)$$

This mathematical restriction, originally introduced for general-purpose manipulation detection [5], forces the kernel to predict the center pixel based exclusively on its immediate neighbors, outputting only the prediction error and acting as trainable high-pass filter. The authors demonstrated that combining all three techniques, rather than relying on any single one, optimally exposes the subtle boundary artifacts of an inpainted region.

Following this enhancement stage, an extraction block derives high-level discriminative features from the resulting noise residual. Instead of relying on a hand-designed backbone such as ResNet, this stage utilizes a highly optimized convolutional cell structure automatically discovered through Neural Architecture Search (NAS).

Finally, a decision block aggregates these extracted features to produce the per-pixel prediction mask. It utilizes a dual attention mechanism: a global attention module reduces intra-class variance by grouping similar features across the entire image, while a local attention module enforces spatial consistency within small neighborhoods, ensuring coherent and precise detection boundaries.

In the course of this work, we attempted to use IID-Net as a candidate baseline for comparison, but found that its Neural Architecture Search stage requires multi-GPU parallel training that could not be accommodated on the single consumer-grade GPU used throughout this thesis, see Chapter 5; this practical obstacle is itself a small, concrete illustration of the accessibility gap that motivates the approach proposed in this work.

A more recent example of this family combines a residual convolutional backbone with a feature pyramid, which aggregates information across multiple spatial scales before the final prediction, specifically to reduce the variability in appearance and size of inpainted regions and to improve consistency across different generative engines [1]. This family of methods generally achieves strong performance when tested on manipulations produced by the same inpainting engines seen during training, since the network can specialize in the exact artifact distribution of those methods, but, mirroring a pattern already seen in Section 2.2, this specialization frequently comes at the cost of weaker generalization to inpainting engines never encountered during training, a limitation examined at length for the method proposed in this thesis in Chapter 5.5.

2.3.4 The State of the Art: Multi-Modal Fusion Networks

The absolute state of the art in forgery localization has recently shifted toward massive, multi-modal architectures that fuse global semantic understanding with low-level signal processing. Modern generative inpainting models produce textures that are perfectly coherent in the RGB space and increasingly sophisticated at faking natural noise. Consequently, looking at the image through a single domain is no longer sufficient.

State-of-the-art frameworks address this by combining global semantic understanding with low-level signal processing, though the specific fusion strategy varies across architectures. TruFor [19], illustrated in Figure 2.5, processes the raw RGB image through a Transformer-based backbone to extract global semantic features, while simultaneously extracting a Noiseprint++ residual, an image-independent camera fingerprint of the kind discussed in Section 2.3, through a dedicated CNN branch; the two representations are then combined through a series of cross-attention layers, allowing the semantic branch

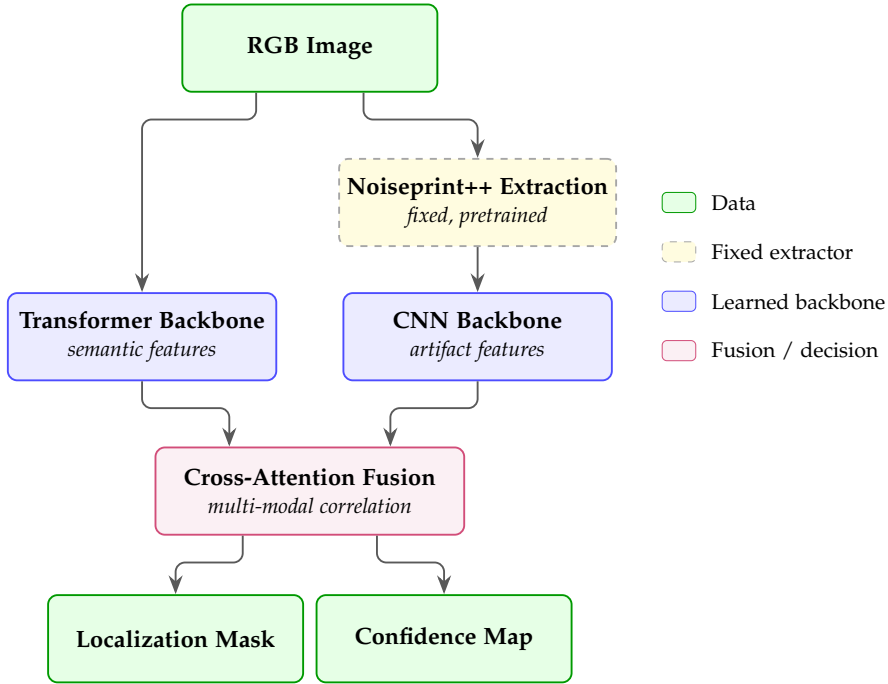


Figure 2.5: TruFor’s architecture [19]. A fixed Noiseprint++ extractor derives a camera fingerprint from the RGB image; the semantic and artifact branches are fused via cross-attention into a mask and a confidence map.

to specifically query the noise branch for local inconsistencies. A related but architecturally distinct approach, ObjectFormer [44], instead extracts high-frequency features directly from the RGB image and combines them with standard RGB features as a single, joint multimodal patch embedding fed to a transformer; rather than fusing two independent backbones, it introduces a set of learnable object prototypes that model object-level consistency across different regions of the image, refining patch embeddings to capture the boundary-level inconsistencies typical of manipulated objects.

Furthermore, TruFor architecture includes a dedicated confidence-estimation branch, explicitly trained to output a map indicating how reliable its own prediction is for any given region, acknowledging that no learned fingerprint generalizes perfectly to every manipulation type. While these multi-branch, fusion-based transformers achieve incredible performance metrics, they require intensive computational resources for both training and inference. They represent the current peak of detection capability but are fundamentally inaccessible for rapid fine-tuning by independent researchers or journalists facing new, emerging generative threats.

2.3.5 Foundation Models for inpainting localization

The accessibility bottleneck of multi-modal fusion networks directly motivates the core methodology of this thesis. Rather than building massive parallel architectures, a recent and highly promising paradigm leverages the latent power of self-supervised Vision Foundation Models.

These models are trained on vast amounts of unlabelled data to understand the deep structural and semantic rules of natural images. Research indicates that representations learned by strong pretrained backbones, such as DINOv3, contain sufficient intrinsic information to identify generative anomalies without requiring any specialized noise-extraction branches. To exploit this, the massive backbone is kept entirely frozen. The forensic task

is reduced to training a lightweight classification head that projects the high-dimensional spatial tokens into a binary prediction mask [14].

Mathematically, if the frozen backbone produces a set of spatial patch embeddings $f_\theta(x)$, the localization probability y for a specific patch can be obtained through a simple projection:

$$y = \sigma(W \cdot f_\theta(x) + b), \quad (2.9)$$

where W and b are the only trainable parameters, and σ is the sigmoid activation function.

This frozen-backbone paradigm isolates the semantic capabilities of the network, answering a specific research question: Can general-purpose self-supervised visual representations contain enough information to localize inpainting artifacts without an explicitly designed forensic feature extractor? Most importantly, this architecture can be fully trained and deployed on a single consumer-grade GPU. This creates a highly agile, reproducible baseline that allows smaller laboratories to quickly fine-tune the detection head against newly released generative models, maintaining relevance in a fast-paced adversarial landscape.

A concurrent and directly relevant development further refines this frozen-backbone paradigm using Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique that, instead of updating the backbone’s original weights directly, injects a small trainable low-rank update alongside each frozen weight matrix, adapting the pretrained representations slightly without the cost of full fine-tuning. Applied to a DINOv3 backbone paired with a lightweight convolutional decoder, this approach was recently shown to substantially outperform both full fine-tuning and a wide range of specialized detectors on general-purpose manipulation localization benchmarks, using only a small fraction of the backbone’s total parameters as trainable [52]. This result suggests that allowing a small, efficient adaptation of the backbone’s internal representations, rather than freezing it completely as done in this thesis, may recover additional performance without abandoning the accessibility benefits that motivate the frozen-backbone approach, an interesting direction for future extensions of the method proposed here.

2.3.6 Toward Content-Aware Region Modeling: An Unexplored Direction

A structural limitation shared by most of the methods discussed in this section, including the foundation model approach proposed in this thesis, is the reliance on rigid geometric grids. Whether an image is divided into fixed blocks for correlation analysis or square patches for Vision Transformer tokenization, the image is partitioned using rules that completely ignore the actual visual structure of the scene.

Because inpainting algorithms manipulate entire semantic objects rather than arbitrary square grids, a rigid bounding box often fragments a manipulated object across several unrelated tokens. This weakens the coherence of the extracted features and blurs the predicted detection boundaries. A recent advancement addresses this structural flaw by replacing fixed grids with content-coherent, irregularly shaped regions [46], as shown in Figure 2.6. Utilizing a differentiable clustering procedure inspired by superpixel segmentation, the network dynamically groups feature elements based on spatial and semantic similarity. These content-aware regions serve as the nodes of a graph neural network, allowing the model to propagate forensic clues across the exact shape of the manipulated object rather than bleeding into the authentic background.

Since the DINOv3 architecture employed during this work inherently tokenizes the input into a fixed square grid before the classification head processes the data, it inherits this geometric limitation. The replacement of this fixed tokenization with a content-aware pooling layer before the classification head represents a promising architectural upgrade worth investigating in future research.

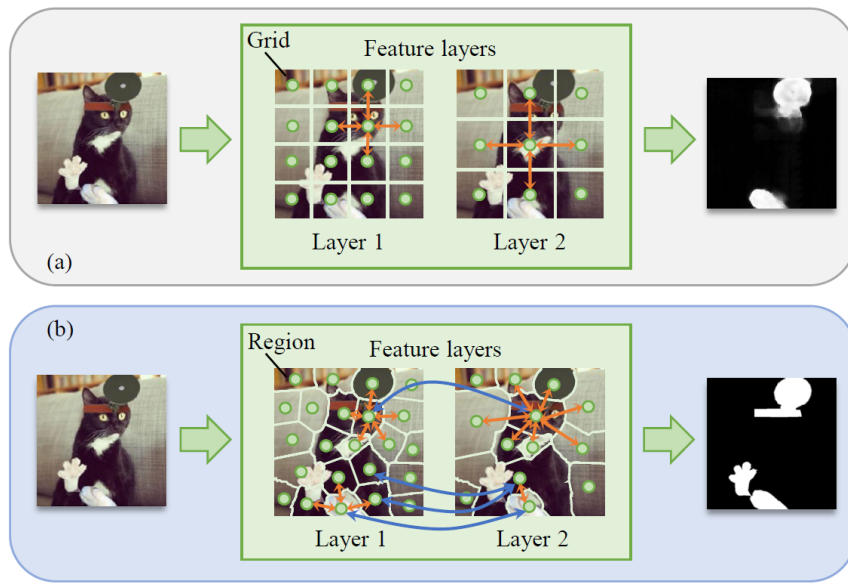


Figure 2.6: Grid-based (a) versus content-coherent region-based (b) correlation modelling. Reproduced from Wang et al. [46].

2.3.7 Key takeaways and Approach Motivations

Together, the methods reviewed in this section reveal two recurring limitations that directly motivate the design of this thesis.

First, both classical and deep-learning-based detectors, including several of the inpainting-specific and multi-modal approaches discussed above, are typically developed and evaluated primarily on manipulations produced by the same generative engines seen during training. Their ability to generalize to unseen, more advanced generators is comparatively rarely tested, an issue that inspired us to perform extensive out-of-distribution testing on manipulations produced by engines absent from the training set, discussed in Chapter 5.5.

Second, forensic state-of-the-art architectures are generally very focused on performance, assuming access to substantial amounts of data and computational resources for training and, in several cases, for inference as well. These requirements make heavier models more difficult and costly to retrain, while the foundation-based approach explored in this work relies on a frozen-backbone providing general image knowledge; it can be adapted to newer inpainting generators by just training the lightweight classification head, requiring only a limited amount of resources and initial data to obtain the promising results.

Chapter 3

Dataset selection and inpainting generation

3.1 The Challenge of Data Collection for Inpainting Detection

To address the problem of inpainting detection and localization, we need to train and evaluate our models using a very specific type of data: images which have been partially modified by some algorithm or generative model, paired with the exact ground-truth binary mask indicating the modified region. Obtaining this type of data at scale presents several difficulties.

The first challenge we face is the absence of a universally adopted standard benchmark for the task. In similar tasks such as image classification or object detection, some datasets like ImageNet or MS-COCO have become the de facto standard, but since inpainting detection is relatively new and rapidly evolving, available datasets are scarce and automatic inpainting procedures are still experimental. Consequently, generating high-quality inpainted images is an expensive task in terms of both time and resources. However, the absence of an industry standard leads many authors to create and use their own small-scale datasets of inpainted images. It is important to note that generative techniques are in constant and rapid evolution, which means that the models used to create these local datasets can become outdated very quickly compared to the state of the art.

A second critical problem is the quality of the inpaintings: a dataset composed of low-quality inpaintings, where manipulations are easily visible even at first glance, is not useful to train models able to detect the modifications created using modern state-of-the-art generative models, which produce results increasingly similar to reality and difficult to detect. The necessity of high-quality inpainted images for the training therefore further restricts the availability of datasets suitable for the task.

Finally, we require ground-truth binary masks of the modified region to train our models. This means that, regardless of the inpainting procedure (manual, semi-automatic, or automatic), these masks must be provided; unfortunately, this is not always the case. Many modern state-of-the-art models regenerate the entire image from scratch every time, so they do not provide a binary mask indicating the locally modified area. Furthermore, a significant number of these top-tier models are closed-source, accessible exclusively through paid APIs, and do not allow users to inspect or control crucial generation parameters. This lack of transparency and reproducibility severely limits their adoption in rigorous scientific applications. Since no ground-truth binary mask is available for these methods, we excluded all the datasets created in this manner.

In this work, we utilize various sources for our datasets, which are described in the following sections. In particular, we use three distinct datasets for in-distribution training

and evaluation, and out-of-distribution (OOD) testing, i.e., evaluating the model’s generalization capabilities on images generated differently from those used during training.

3.2 Automatic inpainted images generation: The Beyond the Brush pipeline

A concrete approach for the automated construction of inpainting datasets has been proposed by Bertazzini et al. in the Beyond the Brush [8] paper, which introduces an automated pipeline for the generation of realistic inpainted images starting from pristine ones. The framework operates through three main stages:

1. **Mask Extraction:** Meaningful regions to be inpainted are extracted from the source image through a sequential three step segmentation procedure. First, the Recognize Anything Model (RAM) [55] processes the image and generates a list of tags identifying the main objects. Second, Grounding DINO [30], a Transformer-based open-set detector, takes these tags as text input and generates bounding boxes with confidence scores for each object. Third, the Segment Anything Model (SAM) [24] converts each bounding box into a precise pixel-level binary mask. For each image, the three masks with the highest confidence scores are retained and categorised by size: small (0 to 15% of the image area), medium (15 to 30%), and large (30 to 60%). Finally, a morphological opening followed by a closing operation is applied to each mask to reduce noise at the object boundaries.
2. **Prompt Generation:** A Visual Language Model (VLM) generates inpainting prompts for each masked region. The image is presented to the VLM with the selected region highlighted by a green bounding box, and the model is asked to describe a semantically coherent replacement for the object within the box. Following the findings of Wei et al. [47] on chain-of-thought prompting, the VLM is also asked to explain its reasoning, which improves the quality and coherence of the generated descriptions. In the original pipeline, the authors used Mini InternVL Chat 4B [10] as the VLM to generate five distinct prompts per mask to account for occasional inconsistent responses.
3. **Inpainting Generation:** The inpainted images are generated using Fooocus [54], an open-source system built on Stable Diffusion, accessed through its REST API. Given the source image, the binary mask, and one of the generated prompts, Fooocus fills the masked region with new content using default parameter settings and a fixed seed, ensuring reproducibility. Each combination of mask and prompt produces one inpainted image, thus a single source image yields multiple inpainted variants.

3.3 Our Generated Dataset: BtB-COCO

We adopted the Beyond the Brush pipeline, applying a few modifications, to construct a dataset of generatively inpainted images using a subset of pristine images taken from the MS-COCO 2017 [28] validation set.

3.3.1 Source Image Selection

We created a script to select 250 images from the COCO validation set applying a simple quality filter: we only included images containing a minimum of three annotated objects in COCO segmentation. This filter guarantees each image contains enough distinct subjects to use as candidates for the inpainting pipeline, reducing the probability of generating

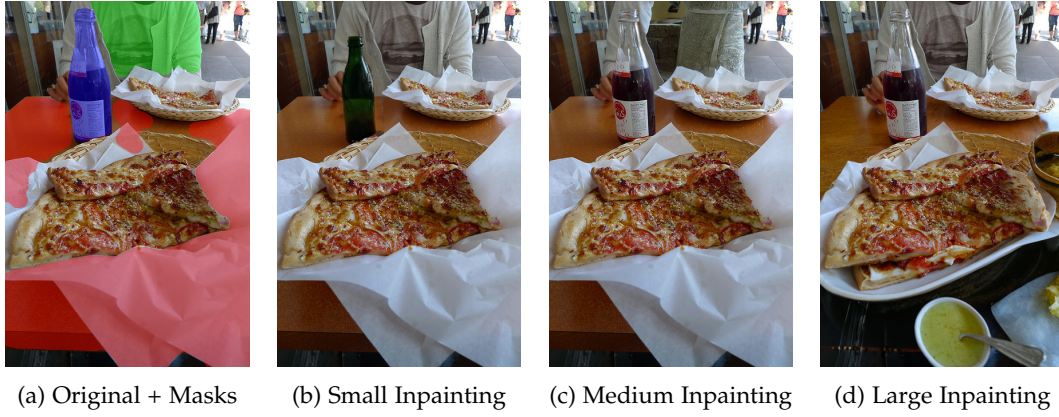


Figure 3.1: On the left: the original image overlaid with the three independent binary masks extracted by the pipeline (blue for small, green for medium, red for large) for illustrative purposes. On the right: the three corresponding inpainted variants generated by Fooocus.

excessively small masks, which would carry little significance. We chose the MS-COCO dataset because it is widely used for similar tasks and provides object-level segmentation annotations that allow us to select suitable images through a simple filtering criterion. Moreover, its moderate-resolution images are well suited to our computational constraints. In addition to the selected 250 images, we added 4 images captured on our personal camera to test how the pipeline would behave on higher resolution images.

3.3.2 The Renewed Pipeline: Beyond the Brush Modifications

The execution of the original Beyond the Brush pipeline on our local hardware highlighted several computational bottlenecks and inefficiencies. To adapt the framework to our resources and improve the overall quality of the dataset, we introduced several key modifications to the pipeline, which is shown in Figure 3.2.

During the segmentation stage, generating a large number of bounding boxes frequently caused Out-of-Memory (OOM) errors. To address this, we used a smaller batch size and we modified the script to retain only a maximum of three masks per image, effectively solving the memory constraints. Furthermore, the morphological opening and closing operations used to smooth the masks occasionally eroded extremely thin objects entirely, resulting in completely black binary masks. We implemented an auxiliary filtering script to detect and remove these empty masks before the prompt generation stage, preventing the waste of computational resources on invalid regions.

Regarding the prompt generation, the original pipeline employed Mini-InternVL-Chat-4B as the Visual Language Model (VLM). This model proved to be computationally heavy and slow for our hardware setup. We replaced it with Qwen2-VL-2B-Instruct [45], a highly efficient 2 billion parameter model capable of running in 8 bit quantization. To compensate for the reduced parameter count and ensure high-quality outputs, we significantly expanded the text prompt fed to the model. The new prompt explicitly forces the VLM to analyze the room style, lighting, and surrounding objects, enforcing the generation of diverse and contextually coherent replacements. Seeking an optimal trade-off between semantic diversity and generation time, we also reduced the number of generated prompts from five to three per mask.

Finally, we integrated and tested three different inpainting architectures to explore how alternative models for inpainting perform compared to the original one used in the paper. We implemented Fooocus [54] (the model used in the original BtB pipeline, based on Stable Diffusion XL), Stable Diffusion v1.5 [41], and Simple LaMa [43]. It is worth noting that while Fooocus and Stable Diffusion are diffusion-based models, LaMa is a fast,

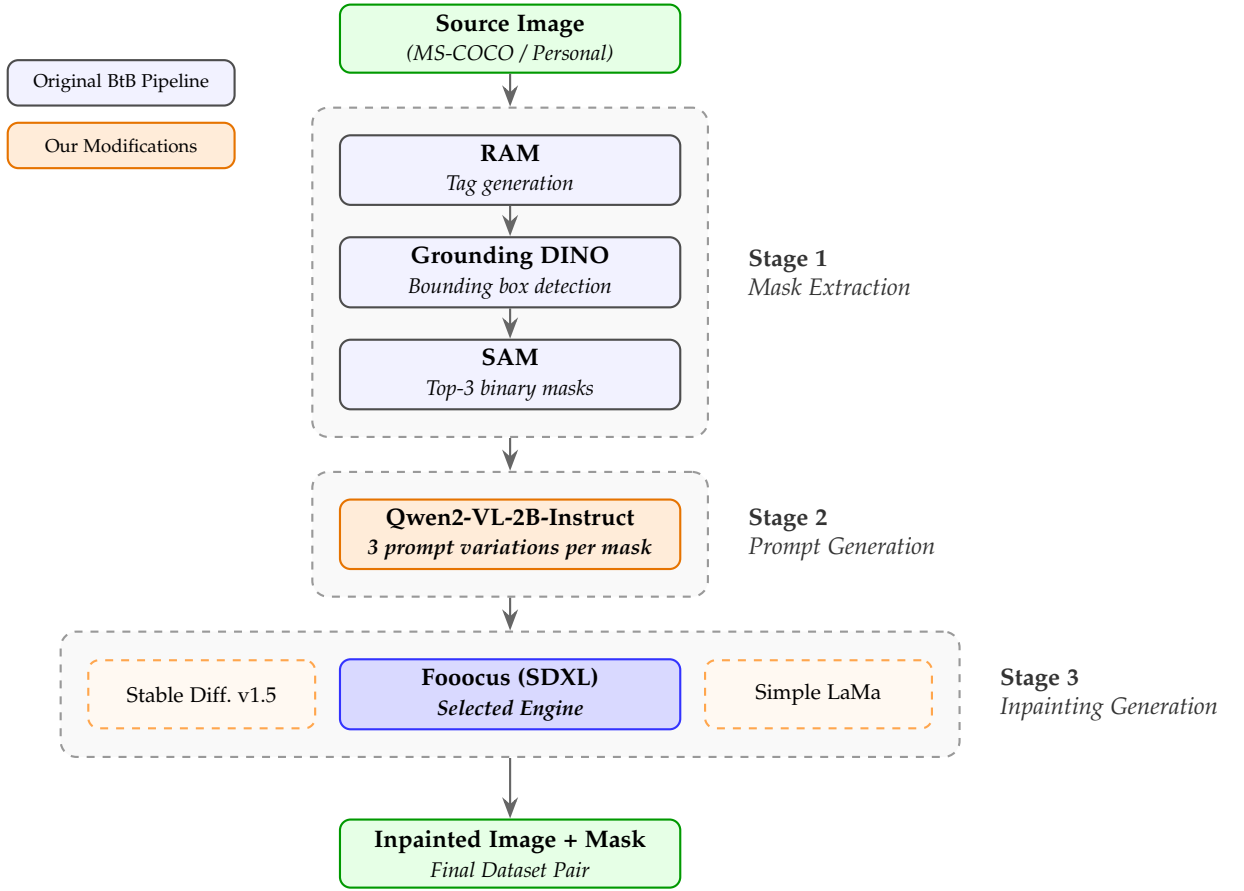


Figure 3.2: The modified Beyond the Brush pipeline adopted in this work. A source image passes through three sequential stages: mask extraction via RAM, Grounding DINO, and SAM; prompt generation via the highly efficient Qwen2-VL-2B-Instruct; and inpainting generation. Among the three inpainting engines tested, Foococus was selected for its superior photorealism.

lightweight feed-forward network based on Fast Fourier Convolutions (FFCs). Although LaMa proved to be significantly faster, the visual quality and realism of its outputs were remarkably inferior when inserting new complex semantics. Consequently, we selected Foococus as our definitive inpainting engine. Despite being more resource-intensive and requiring an external API server, its SDXL-based architecture provided vastly superior blending and photorealism. To support long generation sessions, we also developed a cleanup script to automatically handle occasional API timeouts or server errors.

Applying this refined pipeline to our 254 curated source images, despite being lighter than the original BtB approach, took around 45 hours of runtime on our machines, highlighting the importance of having an industry-standard benchmark to avoid the need of creating personalized datasets. After filtering invalid masks and unsuccessful generations, we obtained a final dataset of 1,423 inpainted images, each paired with its corresponding binary mask. The number of output images is substantially larger than the number of source images because each source image can produce multiple masks, and each valid mask can be associated with multiple inpainting prompts. Figure 3.3 provides a visual comparison of the results produced by the three tested inpainting models given the same source image, mask, and text prompt.



Figure 3.3: Visual comparison of different inpainting backends given the same text prompt and mask. From left to right: Original Image, Binary Mask, Fooocus (SDXL), Stable Diffusion v1.5, and Simple LaMa. Fooocus demonstrates superior capability in generating semantically complex and realistic replacements.

Property	Value
Source images	250 MS-COCO 2017 val + 4 personal photos
Total inpainted images	1,423
Inpainting model	Fooocus (Stable Diffusion XL)
Usage	Out-of-distribution testing

Table 3.1: Statistics of the generated BtB-COCO dataset.

3.4 The Beyond the Brush-Flickr30k Dataset

In addition to our generated dataset, we utilized the public collection released by the Beyond the Brush authors, available on Hugging Face¹. The complete BtB collection is composed of tampered images derived from three different source datasets: Flickr30k, VISION, and FloreView [8]. However, for our experiments, we restricted our download strictly to the BtB-Flickr30k subset. This decision was driven by severe hardware and storage constraints: the VISION and FloreView subsets contain high-resolution pictures (up to 8000×6000 pixels), totaling approximately 190 GB of data [8]. In contrast, the Flickr30k subset consists of lower-resolution images, requiring only about 2.5 GB of storage, which perfectly suited our computational resources.

The BtB-Flickr30k dataset comprises approximately 7,400 images generated using the Fooocus engine [8]. The dataset is evenly balanced among three categories based on the dimension of the inpainting mask: small, medium, and large.

Initially, before settling on SAGI-D as our main training dataset, we used BtB-Flickr30k for our preliminary training and model experiments. To facilitate this, we developed a custom preprocessing script to download and partition the data into training, validation, and test sets using an 80-10-10 split ratio. Since the BtB pipeline produces multiple inpainted variants from a single pristine image, a naive random split would have caused severe data leakage, placing very similar images in both the training and validation/test sets. To prevent this, our script grouped all images by their original source ID, guaranteeing that all variants derived from the same base image were assigned to the exact same split.

Despite these rigorous precautions, empirical tests revealed that the limited size of the data set led to rapid model overfitting. Consequently, we opted to use a much larger dataset, SAGI-D (described in Section 3.5), for the definitive training phase. The BtB-Flickr30k dataset was therefore repurposed for rapid prototyping of new models and, most importantly, as a massive Out-of-Distribution (OOD) benchmark. Since the final models are trained exclusively on SAGI-D, the entirety of the BtB-Flickr30k dataset acts

¹<https://huggingface.co/datasets/lesc-unifi/beyond-the-brush>

as completely unseen data, providing a robust evaluation of the models’ generalization capabilities on images generated via Fooocus.

3.5 The SAGI-D Dataset

To overcome the size limitations and the resulting overfitting issues encountered with the BtB-Flickr30k dataset, we selected the SAGI-D dataset for the primary training and evaluation of our models. SAGI-D is a large-scale collection specifically designed for the detection and localization of generative image manipulations.

For our experiments, we curated a subset of approximately 15,000 images, totaling around 24 GB of data. During the extraction process, we strictly filtered the dataset to include only “semantically preserved” (*sp*) images, where we have a mask of the manipulated area and the pixels outside it remain unaltered. This filtering criterion is crucial for the task of inpainting localization, as in our approach we use the ground truth area of the manipulation during the training to enable the localization of the inpainting. Consequently, “fully regenerated” (*fr*) instances were explicitly excluded from our subset.

The final selected subset features a balanced distribution of images modified by five modern diffusion-based inpainting architectures: RemoveAnything [53], HDPainter [35], BrushNet [23], InpaintAnything [53], and PowerPaint [56]. A detailed technical description of each of these five architectures, together with Fooocus, the engine used for out-of-distribution evaluation, is provided in Section 3.6. To ensure the integrity of the ground truth during the training phase, we implemented a rigorous quality control step. Specifically, we verified the dimensional alignment between every generated image and its corresponding binary mask. We identified and removed a small number of pairs, primarily generated by the HDPainter model, where the image and mask dimensions did not match. This precaution prevented potential tensor shape mismatches and training crashes, resulting in a final, perfectly aligned dataset of 14,796 images.

Finally, we partitioned the curated SAGI-D dataset into three main subsets using a stratified splitting strategy based on the specific inpainting model, ensuring a balanced representation across all phases:

- **Training Set (80%):** used to optimize the neural network weights and teach the model to localize the inpainted boundaries;
- **Validation Set (10%):** employed during the training loop for hyperparameter tuning, overfitting detection and early stopping supervision;
- **Test Set (10%):** reserved exclusively for the definitive in-distribution evaluation of our final models.

This precise subdivision guarantees that each split contains an equal proportion of images generated by each of the five selected architectures, providing a robust foundation for our detection networks.

3.6 Inpainting Generators Used in This Thesis

As discussed in Chapter 2, due to their advanced generative capabilities, LDMs dictate the current standard in digital manipulation. For this reason, the experimental datasets curated and evaluated in this thesis, namely SAGI-D, Flickr30k and BtB-COCO, are exclusively populated by manipulations generated through modern diffusion models. Rather than relying on a single architecture, the forensic framework proposed in this work is explicitly tested against a diverse set of modern inpainting engines to guarantee robust generalization:

- **BrushNet:** The common approach of feeding the masked image to the U-Net as extra input channels forces a single set of shared weights to jointly handle both background preservation and content generation. Brushnet instead decomposes the network into two parallel branches, one of which is entirely dedicated to processing and preserving background features, injecting this information into the main generative branch at multiple layers, reducing boundary artifacts that arise when both concerns are entangled in a single pathway [23], and creating highly consistent boundary transitions.
- **PowerPaint:** The focus of this model is on versatility rather than fidelity: rather than training separate models for different editing goals, such as removing an object versus inserting a new one, it learns a small set of dedicated task tokens that condition a single shared model to behave differently depending on which token is provided, allowing one network to serve multiple distinct inpainting objectives [56].
- **HD-Painter:** Standard inpainting diffusion models tend to prioritize blending smoothly with the surrounding texture over strictly following the user’s text prompt, especially at high resolution; HD-Painter introduces a specialized attention mechanism designed to keep the generation faithful to the prompt even when this conflicts with the easiest, blend-friendlier solution [35].
- **InpaintAnything / RemoveAnything:** Both architectures build on the same underlying pipeline, combining the Segment Anything Model, an interactive segmentation network capable of producing a precise object mask from a single click, with a diffusion-based generative backbone, but they use this combination for two different purposes. RemoveAnything is designed purely for object removal: once the target object is segmented, an inpainting model reconstructs the background as if the object had never been there, guided only by the surrounding context rather than any text description, producing a clean, semantically preserved reconstruction. InpaintAnything, by contrast, additionally accepts a user-provided text prompt describing new content to generate in place of, or around, the segmented region, turning the same segmentation-driven mask selection mechanism into a tool for content replacement rather than pure removal. Both pipelines share the practical advantage of automating what was previously a manual and error-prone step, precise mask selection, allowing high-quality inpainting without requiring the user to possess any image-editing skill [53]
- **Foocus:** It is an open-source interface built on top of Stable Diffusion XL that automates prompt engineering internally, producing polished, aesthetically coherent results with comparatively little effort required from the user [54]; it was used in this thesis to build the out-of-distribution datasets described in Section 3.4 and Section 3.3. This engine is entirely absent from the training set, which makes it a meaningful test of generalization discussed in Chapter 5.5.

3.7 Dataset Selection and Summary

As discussed in the previous sections, the selection of the datasets was driven by the need for a large volume of high-quality, masked inpaintings to avoid overfitting during the training phase. For this reason, SAGI-D was selected as our primary dataset for in-distribution training, validation, and testing. Conversely, to rigorously evaluate the generalization capabilities of our models in real-world scenarios, we employed both the BtB-Flickr30k dataset and our custom BtB-COCO dataset exclusively as Out-of-Distribution (OOD) benchmarks. This ensures that the models are tested on images generated with different semantics and via a different inpainting engine (Foocus) than those seen during training.

Table 3.2 summarises the main characteristics of the three datasets used in this work and the role of each in the experimental pipeline.

Dataset	Source	Inpainting model	Size	Usage
SAGI-D	COCO + RAISE	BrushNet	~14,800	Train / Val / Test
		HDPainter		
		PowerPaint		
		RemoveAnything		
		InpaintAnything		
BtB-Flickr30k	Flickr30k	Fooocus (SDXL)	~7,400	OOD Test
BTB-COCO (ours)	COCO val + personal	Fooocus (SDXL)	~1,400	OOD Test

Table 3.2: Summary of the datasets used in this work. OOD indicates out-of-distribution evaluation, meaning the model is tested on images generated with methods not seen during training.

Chapter 4

The Dinolizer Approach

4.1 Overview

In recent years, the development of large neural networks trained on enormous quantities of images has produced models capable of extracting extremely rich and general visual representations from any image. These models, known as Vision Foundation Models, have been trained on hundreds of millions of images and have learned to recognize and encode a vast range of visual patterns, textures, shapes, and structures. Rather than training a new complex model from scratch for each specific task, a common and effective strategy is to use one of these pretrained models as a fixed feature extractor, and to train only a small, lightweight component on top of it for the task at hand.

This is precisely the approach followed in this work, illustrated in Figure 4.1. We use DINOv3, a Vision Foundation Model, as a frozen feature extractor. The model is never modified during our training process: its weights are fixed, and it is used exclusively to transform input images into rich numerical representations. On top of these representations, we train a small classification component, which we call the classification head, whose sole purpose is to decide, for each region of the image, whether that region has been modified by a generative inpainting process or not.

As will be explained in detail in the following sections, the foundation model internally divides each image into a regular grid of small spatial regions and produces one numerical representation vector for each region. The classification head then operates on these vectors, assigning to each region a probability of having been inpainted. The resulting grid of predictions is then scaled back to the original image resolution through interpolation, producing a dense heatmap where each pixel carries a value between 0 and 1 indicating the estimated probability of manipulation.

The main advantage of this design is its computational efficiency. Since the heavy feature extraction work is delegated entirely to the pretrained model, only the small classification head needs to be trained. This dramatically reduces the amount of data, time, and computational resources required compared to training a full model from scratch, while still achieving strong localization performance.

4.2 The DINOv3 Backbone

4.2.1 Vision Transformers

To understand how DINOv3 processes images, it is necessary to first introduce the concept of a Vision Transformer (ViT) [15]. A Vision Transformer is a type of neural network that processes images in a fundamentally different way from traditional convolutional neural networks (CNNs). Instead of scanning the image with small local filters as CNNs do, a ViT

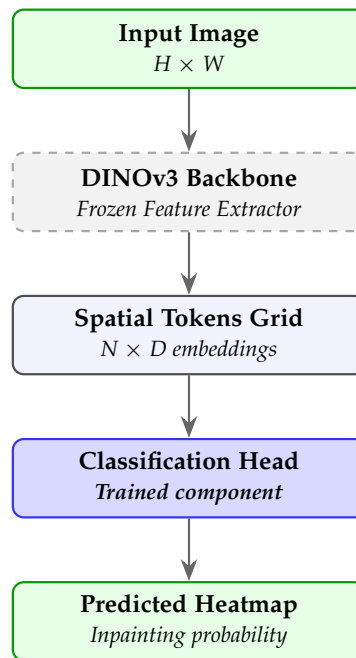


Figure 4.1: Overview of the proposed DinoLizer pipeline. An input image is processed by the frozen DINOv3 backbone to extract a dense grid of spatial embeddings. These representations are then fed into a lightweight classification head, which is the only component updated during training, to produce a pixel-level forgery heatmap.

begins by dividing the input image into a grid of non-overlapping square regions called patches.

Each patch is then flattened into a one-dimensional embedding, which lives in a space of dimension D . We then feed our sequence of patch embeddings to a transformer, in the same way we would do with word embeddings in a regular transformer.

The transformer processes the sequence through a series of layers, each of which applies a mechanism called self-attention, which allows every patch in the sequence to compare itself with every other patch and to incorporate information from the patches it finds most relevant. This is the key property that distinguishes ViTs from CNNs: a convolutional filter can only look at a small local neighbourhood around each pixel, whereas self-attention allows each patch to attend to any other patch in the image regardless of distance. This global reasoning capability makes ViTs particularly well suited for tasks that require understanding the relationship between distant parts of an image.

In addition to the patch embeddings, ViTs typically include one or more special tokens in the sequence. The most common is the class token, often written as [CLS], which does not correspond to any specific image region but aggregates global information about the entire image and is used for image-level classification tasks. More recent variants also include register tokens [13], which serve as additional memory slots that help the model distribute computation more effectively across the sequence without polluting the patch embeddings with global information.

4.2.2 Self-Supervised Learning

The DINO pretraining framework does not require manually annotated labels. Instead, it makes use of a process called self-supervised learning, in which the model learns its representations by solving tasks that can be automatically constructed from the training data itself, without any human annotation.

The specific technique used by the DINO family of models [38] is based on self-distillation. Two copies of the same network are maintained during training: a student network and a teacher network. Given an image, two different augmented versions of it are created (for example, two random crops with different colour adjustments). The student sees one version and the teacher sees the other. The student is trained to produce output representations that match those of the teacher, even though the two networks saw different views of the same image. The teacher is not trained directly through gradient descent but is instead updated as a slow moving average of the student weights. Through this process, both networks gradually learn representations that are consistent and invariant to superficial image transformations, capturing instead the deeper structural and semantic content of the image.

The result is a model that, despite having never seen a single human-provided label, produces high quality representations that are useful for a wide range of downstream tasks including classification, segmentation, and, as demonstrated in this work, forgery localization.

4.2.3 DINOv3 ViT-S/16

The first backbone variant used in this work is DINOv3 ViT-S/16 [42], a Vision Transformer with a patch size of 16×16 pixels. Given an input image of resolution $H \times W$ pixels, assuming H and W are divisible by the patch size, the model divides it into a regular grid of patches, where the total number of patches is:

$$N = \frac{H \times W}{16 \times 16}. \quad (4.1)$$

Each patch is encoded into an embedding vector of dimension $D = 384$. The model also prepends one [CLS] token and four register tokens to the sequence before processing, for a total of $N + 5$ tokens in each sequence. After the transformer layers have processed the full sequence, we discard the [CLS] and register tokens and retain only the N patch embeddings. Each of these embeddings encodes the visual content of its corresponding 16×16 pixel region of the original image.

For a 320×320 input image, this produces $N = 400$ patch embeddings arranged in a 20×20 spatial grid. These 400 vectors, each of dimension 384, form the input to the classification head described in Section 4.3.

4.2.4 DINOv3 ConvNeXt (Tiny and Small)

The second and third backbone variants both belong to the ConvNeXt family [31], a type of convolutional neural network redesigned to incorporate many of the design principles that made Vision Transformers successful, while retaining the computational structure of convolutional networks.

ConvNeXt processes the image in two phases. In the first phase, called the stem, a single convolution transforms the raw input image into an initial feature map with a reduced spatial resolution and an increased number of channels. Specifically, for a 640×640 input, the stem produces a feature map of spatial size 160×160 with 96 channels. This initial step can be thought of as the network’s first compact encoding of the image.

In the second phase, the feature map is processed through four successive stages. Between each pair of stages, a downsampling operation halves the spatial resolution and doubles the number of channels, so the representation becomes progressively more abstract and compact as it moves deeper into the network. For the Tiny and Small variants evaluated in this work, the progression of the channel through the four stages is $96 \rightarrow 192 \rightarrow 384 \rightarrow 768$, producing a final feature map of size 20×20 with 768 channels for a 512×512 input.

Within each stage, the network applies a series of ConvNeXt blocks. Each block works in three steps. First, a depthwise convolution with a 7×7 kernel is applied: each channel

of the feature map is convolved independently with a 7×7 filter, so each spatial position gathers information from a 7×7 neighbourhood around itself without mixing channels. Second, a pointwise 1×1 convolution mixes the information across channels at each spatial position independently. Third, another pointwise convolution projects the result back to the original channel dimension. A normalization step and GELU activation function are also applied within each block.

The 7×7 kernel is significantly larger than the 3×3 kernels used in most classical CNN architectures, and this design choice was directly inspired by the large receptive fields of transformer self-attention. However, the fundamental difference from a ViT remains: in a ViT, every patch can directly attend to every other patch in the image in a single operation. In ConvNeXt, each position only processes a local neighbourhood at each block, and global context is built up gradually through the stacking of many blocks across multiple stages. By the final stage, each position has an effective receptive field that covers a large portion of the original image, but this coverage is achieved progressively rather than all at once.

To ensure that both the ViT and ConvNeXt backbones can seamlessly interface with our classification heads, DINOv3 wraps the ConvNeXt output into a sequence of spatial tokens. For a 320×320 input, the effective patch size is 32×32 pixels, yielding a spatial grid of $10 \times 10 = 100$ tokens, each with dimension $D = 768$. This grid is coarser than the 20×20 grid produced by the ViT backbone for the same input, which means the localization heatmap will have a lower spatial resolution before upsampling. To compensate for this, experiments with the ConvNeXt backbone use a larger input resolution of 640×640 pixels, which produces a 20×20 spatial grid, comparable to the ViT configuration.

We evaluate two variants of DINOv3 ConvNeXt [42]: ConvNeXt Tiny (29M parameters) and ConvNeXt Small (50M parameters). Both share the same channel progression, embedding dimension, and effective patch size. The only structural difference is the number of blocks per stage: Tiny uses a depth configuration of [3, 3, 9, 3], while Small uses [3, 3, 27, 3], making Small a strictly deeper network with greater representational capacity at the cost of increased computation.

In all backbone configurations, the model weights are kept completely frozen throughout the entire training process. No gradient is ever computed with respect to the backbone parameters. Only the classification head, described in the next section, is updated during training.

4.3 Classification Heads

After the backbone produces its spatial token sequence, a classification head is responsible for mapping each token to a prediction. The head receives as input the N spatial tokens $\mathbf{z}^{(L)} = [z_1^{(L)}, \dots, z_N^{(L)}] \in \mathbb{R}^{N \times D}$ and must produce for each token i a scalar logit $\ell_i \in \mathbb{R}$. A logit is simply an unconstrained real number: positive values indicate the model believes that region is inpainted, negative values indicate it believes the region is authentic. The logits are then passed through a sigmoid function, which maps any real number to the interval $(0, 1)$, producing the per-region inpainting probability. These probabilities are arranged back into the spatial grid and upsampled to the original image resolution through bilinear interpolation.

We design and evaluate three different head architectures, illustrated in Figure 4.2 and described below.

4.3.1 Linear Head (DinoLizer)

The simplest possible head is a single linear transformation applied independently to each token:

$$\ell_i = W \cdot z_i^{(L)} + b, \quad W \in \mathbb{R}^{1 \times D}, \quad b \in \mathbb{R}. \quad (4.2)$$

This is a single matrix multiplication followed by the addition of a bias term. The weight matrix W has only D entries, and together with the bias the head has $D + 1$ trainable parameters in total. For $D = 384$ this means only 385 values need to be learned, which is a remarkably small number. Each token is transformed into a scalar completely independently of all other tokens: no information is shared between neighbouring regions at this stage.

This is the approach proposed in the original DinoLizer paper [14], which showed that this minimal head achieves strong localization performance when the backbone representations are sufficiently rich. In our implementation we additionally apply a Dropout regularization layer with probability $p = 0.3$ before the linear projection. Dropout randomly sets a fraction of the input values to zero during each training step, which prevents the head from relying too heavily on any single feature and helps reduce overfitting.

4.3.2 MLP Head (DinoLizerMLP)

The second head variant introduces a non linear transformation by adding a hidden layer between the input token and the output logit:

$$\ell_i = W_2 \cdot \text{GELU}(W_1 \cdot z_i^{(L)} + b_1) + b_2, \quad (4.3)$$

where $W_1 \in \mathbb{R}^{(D/2) \times D}$ first projects the token from dimension D down to dimension $D/2$, a non-linear activation function called GELU [20] is then applied element-wise, and $W_2 \in \mathbb{R}^{1 \times (D/2)}$ projects the result down to the final scalar logit. GELU is a smooth activation function that introduces non-linearity into the computation, allowing the head to learn more complex relationships between the input features and the output prediction than a purely linear transformation could capture.

This architecture is commonly referred to as a two-layer Multi-Layer Perceptron (MLP). The total number of trainable parameters is approximately $D \times D/2 + D/2 + 1$, which for $D = 384$ is roughly 74,000, still negligible in comparison to the backbone. Dropout with $p = 0.3$ is applied before both linear layers.

The DinoLizer paper [14] evaluated a similar MLP head and found only marginal improvements over the linear head, concluding that the backbone representations are already rich enough that a more expressive head does not provide significant benefit. We replicate this comparison in our experimental setting to verify whether the same conclusion holds for the DINOv3 ViT-S backbone and the SAGI-D dataset.

4.3.3 CNN Head (DinoLizerCNN)

The third head variant takes a fundamentally different approach. Rather than treating each token independently, it exploits the spatial arrangement of the token grid to allow neighbouring tokens to communicate with each other before producing the final prediction. To do this, the flat sequence of N tokens is first reshaped into a two dimensional spatial grid of shape $D \times \sqrt{N} \times \sqrt{N}$, which can be thought of as an image with D channels and spatial dimensions equal to the patch grid size. Standard two-dimensional convolutions are then applied to this grid.

The head is organized in three stages:

$$\mathbf{F}_1 = \text{GELU}(\text{Conv}_{1 \times 1}(\mathbf{Z}_{\text{grid}})) \in \mathbb{R}^{64 \times \sqrt{N} \times \sqrt{N}}, \quad (4.4)$$

$$\mathbf{F}_2 = \text{GELU}(\text{Conv}_{3 \times 3}(\mathbf{F}_1)) \in \mathbb{R}^{32 \times \sqrt{N} \times \sqrt{N}}, \quad (4.5)$$

$$\mathbf{L}_{\text{grid}} = \text{Conv}_{1 \times 1}(\mathbf{F}_2) \in \mathbb{R}^{1 \times \sqrt{N} \times \sqrt{N}}. \quad (4.6)$$

The first convolution is a 1×1 convolution, meaning it operates on each spatial position independently and simply reduces the number of channels from D to 64. This is analogous

to the linear head but applied in a convolutional framework. The second convolution uses a 3×3 kernel with padding of 1 pixel, which means each output position receives contributions from itself and its eight immediate neighbours in the grid. This is where spatial context is introduced: the prediction for each patch region is now influenced by the representations of its surrounding regions. The final 1×1 convolution reduces the 32 channels to a single logit per spatial position. Dropout2D with probability $p = 0.3$ is applied after the first and second stages.

The motivation for this design comes from the nature of inpainting artifacts. The boundary between an inpainted region and the authentic background is often where manipulation traces are most visible, and these traces involve local inconsistencies between neighbouring patches. A head that can reason about neighbouring patches simultaneously is therefore better positioned to exploit these boundary cues than a head that treats each patch in isolation.

The total number of trainable parameters in this head is approximately:

$$D \times 64 + 64 \times 32 \times 9 + 32 \approx 43,000, \quad (4.7)$$

for $D = 384$, which remains very small and preserves the computational efficiency of the overall design.

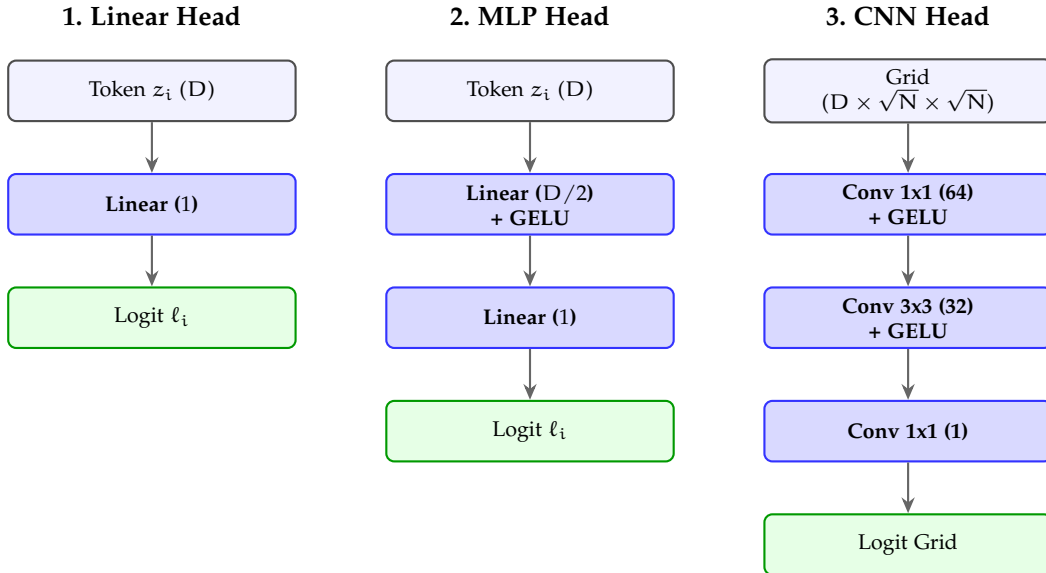


Figure 4.2: Architectural comparison of the proposed classification heads. Linear and MLP heads process spatial tokens independently, while the CNN head reshapes the sequence into a grid to extract local context. (z_i : patch embedding, D : feature dimension, N : total patches, ℓ_i : output logit).

4.4 Adapting the Heads to the ConvNeXt Backbone

The three head architectures described above were designed for the ViT backbone but apply equally to both ConvNeXt variants with two straightforward adjustments.

The first adjustment concerns the embedding dimension. The ViT backbone produces tokens of dimension $D = 384$, while both ConvNeXt backbones produce tokens of dimension $D = 768$. The head architectures are parameterized by D and automatically adapt to whichever value the backbone provides.

The second adjustment concerns the spatial grid size. As described in Section 4.2, both ConvNeXt variants need an image of size 640×640 to produce a 20×20 grid of tokens,

while the ViT produces a grid of the same dimension starting from a 320×320 image. The heads operate on whichever grid size the backbone produces, with no structural changes required.

Since the two ConvNeXt variants are architecturally identical from the perspective of the classification head, applying each of the three heads to each backbone yields six ConvNeXt-based variants. Together with the three ViT-based variants, this brings the total number of architectures evaluated in this work to nine.

Model	Backbone	Head	D	Grid (at 320px)
DinoLizer	DINOv3 ViT-S/16	Linear	384	20×20
DinoLizerMLP	DINOv3 ViT-S/16	MLP	384	20×20
DinoLizerCNN	DINOv3 ViT-S/16	CNN	384	20×20
DinoLizerConvNeXt-T	DINOv3 ConvNeXt Tiny	Linear	768	10×10
DinoLizerConvNeXtMLP-T	DINOv3 ConvNeXt Tiny	MLP	768	10×10
DinoLizerConvNeXtCNN-T	DINOv3 ConvNeXt Tiny	CNN	768	10×10
DinoLizerConvNeXt-S	DINOv3 ConvNeXt Small	Linear	768	10×10
DinoLizerConvNeXtMLP-S	DINOv3 ConvNeXt Small	MLP	768	10×10
DinoLizerConvNeXtCNN-S	DINOv3 ConvNeXt Small	CNN	768	10×10

Table 4.1: Summary of the nine model variants evaluated in this work. The grid resolution refers to the dimension of the spatial token grid produced by the backbone starting from a standard input resolution of: 320×320 . To compensate for the different level of compression of the ViT and ConvNeXt backbones, the trainings made with ConvNeXt use 640×640 input resolution, this way the spatial grid produced has a constant 20×20 dimension across the board.

4.5 Training Pipeline

4.5.1 Data Preparation and Input Modes

The training pipeline supports two distinct input modes, which differ in how images are prepared before being fed to the model.

In **resize mode**, every image in the dataset is rescaled to a fixed square resolution before being passed to the model. This ensures that all images in a training batch have identical dimensions, which allows standard batched processing where multiple images are processed simultaneously. The resize resolution must be a multiple of the backbone patch size (16 for the ViT backbone, 32 for ConvNeXt), so that the image can be evenly divided into a complete grid of patches. In our experiments we use 320×320 pixels for the ViT backbone and 640×640 pixels for the ConvNeXt backbone. The main drawback of this approach is that resizing an image can distort the pixel-level forensic traces that the model needs to detect, particularly if the original image resolution is very different from the target size.

In **patch-based mode**, images are kept at their original resolution and a patch extraction procedure is applied instead. From each image, a fixed number of square crops of size equal to the training window (for example 512×512 pixels) are extracted and used as independent training samples. The extraction is not random: patches are classified into three categories based on the fraction of inpainted pixels they contain, and samples are drawn from each category in controlled proportions. Specifically, boundary patches (those where the inpainted region covers between 20% and 80% of the patch area) are sampled most frequently, as they contain the most informative examples for learning the boundary between authentic and manipulated content. Fully inpainted patches (more than 80% inpainted) and background patches (less than 20% inpainted) are sampled in smaller quan-

tities. This balanced sampling strategy prevents the model from being dominated by the large number of purely authentic patches that would otherwise make up the majority of the training data.

4.5.2 Data Augmentation

During training, a set of data augmentation transformations is applied jointly to each image and its corresponding ground truth mask. Because the mask must always remain aligned with the image, all spatial transformations are applied identically to both.

The augmentation pipeline includes random horizontal and vertical flips, random rotations of up to 15 degrees, random colour jitter (random changes in brightness, contrast, saturation, and hue applied only to the image, not the mask), and random Gaussian blur. Each transformation is applied independently with a configurable probability. Augmentation is applied only during training; during validation and testing, the images are used without any modification. Notably, for each training epoch, different augmentations are randomly applied to the same images, resulting in a substantially larger number of unique training variants over the course of the optimization.

The purpose of augmentation is to expose the model to a wider variety of image appearances during training, reducing the risk that it memorizes specific patterns from the training set rather than learning general features that generalize to new images. This is particularly important given the relatively small size of the SAGI-D dataset compared to the large datasets typically used to train image recognition models.

4.5.3 Loss Functions

The training objective measures how well the predicted heatmap matches the ground truth binary mask. We evaluate two loss function configurations.

Dice Loss

The primary loss function used in this work is the Dice Loss [36], which measures the overlap between the predicted probability map and the ground truth mask. Given the predicted probabilities $p_i \in [0, 1]$ and the binary ground truth labels $g_i \in \{0, 1\}$ for each pixel i , the Dice Loss is defined as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad (4.8)$$

where ϵ is a small constant added to avoid division by zero. The numerator counts the overlap between predictions and ground truth: it is large when the model assigns high probability to pixels that are truly inpainted, and small when it either misses inpainted pixels or incorrectly flags authentic ones. The denominator normalizes by the total predicted and true positive area. When the predicted map perfectly matches the ground truth the Dice Loss equals zero; when there is no overlap at all it equals one.

An important property of the Dice Loss is that it operates on the probability estimates rather than binary predictions, which means gradients can flow continuously through it during backpropagation. This is different from the Dice Score metric used at evaluation time, which operates on binarized predictions obtained by thresholding the probability map at 0.5. The Dice Loss is the baseline loss configuration and corresponds to the approach used in the original DinoLizer paper [14].

Focal Loss Combined with Dice Loss

As an extension, we also experiment with combining the Dice Loss with the Focal Loss [27]. The Focal Loss was originally introduced for object detection tasks where the vast majority of image regions contain no object of interest, creating a severe class imbalance. In our

setting, a similar imbalance exists: most pixels in an image are authentic, and only the inpainted region is manipulated. A standard cross-entropy loss would be dominated by the large number of easy authentic pixels, and the model might learn to simply predict everything as authentic and still achieve a low loss. The Focal Loss addresses this by weighting down the contribution of easy, correctly classified pixels and focusing the learning signal on the hard, ambiguous ones. It is defined as:

$$\mathcal{L}_{\text{Focal}} = -\alpha \sum_i (1 - p_i)^\gamma g_i \log(p_i) - (1 - \alpha) \sum_i p_i^\gamma (1 - g_i) \log(1 - p_i), \quad (4.9)$$

where $\alpha \in [0, 1]$ controls the relative weight given to inpainted versus authentic pixels, and $\gamma \geq 0$ is the focusing parameter that controls how strongly easy examples are down-weighted. When $\gamma = 0$ the Focal Loss reduces to standard cross-entropy. In our experiments we use $\alpha = 0.6$ and $\gamma = 1.5$ as base values for the parameters as this combination yielded the best localization performance in preliminary experiments.

When both losses are active, the combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{Focal}} + \mathcal{L}_{\text{Dice}}. \quad (4.10)$$

EMA Loss Normalization

A practical challenge when combining two losses is that they may operate on very different numerical scales. If one loss is systematically ten times larger than the other, the optimizer will effectively ignore the smaller one, since its gradients will be overwhelmed by those of the larger one. To address this, we optionally apply a normalization strategy based on Exponential Moving Averages (EMA)[40].

An Exponential Moving Average is a running estimate of a quantity that gives more weight to recent observations and less to older ones. It is updated at each step as:

$$\mu_t = \beta \cdot \mu_{t-1} + (1 - \beta) \cdot x_t, \quad (4.11)$$

where μ_t is the current estimate, x_t is the current observed value, and $\beta \in [0, 1]$ is the momentum parameter controlling how quickly the estimate adapts. We use $\beta = 0.98$, which means the estimate changes slowly and is dominated by the long-term average of the loss values.

During training, a separate EMA is maintained for both the Focal Loss and the Dice Loss. At each training step, each loss is divided by its current EMA estimate before being summed:

$$\mathcal{L} = \frac{\mathcal{L}_{\text{Focal}}}{\mu_{\text{Focal}} + \varepsilon} + \frac{\mathcal{L}_{\text{Dice}}}{\mu_{\text{Dice}} + \varepsilon}. \quad (4.12)$$

This keeps each normalized loss close to 1.0 at all times, so their sum remains close to 2.0 regardless of how the absolute magnitudes of the two losses evolve during training. The result is that both losses contribute roughly equally to the gradient updates throughout the entire training process.

4.5.4 Optimizer and Learning Rate Schedule

The optimization is performed using AdamW [32], a variant of the Adam optimizer that adds a weight decay term to the parameter update rule. Weight decay penalizes large parameter values, which discourages the model from overfitting to the training data. We use a weight decay of 10^{-3} and an initial learning rate of 10^{-4} . Crucially, the optimizer is applied only to the parameters of the classification head: the backbone parameters are frozen and never updated.

The learning rate is adjusted during training using a ReduceLROnPlateau scheduler. This scheduler monitors the validation loss and reduces the learning rate by a factor of 0.5

whenever the validation loss has not improved for 4 consecutive validation checks. This allows the optimizer to take smaller and more precise steps as training progresses and the model approaches a good solution.

All forward passes and loss computations are performed in mixed precision using 16 bit floating point arithmetic where possible, with automatic fallback to 32 bit for operations that require higher numerical precision. This reduces memory usage and speeds up training without affecting the quality of the results. A gradient scaler is used to prevent the very small gradient values that can arise in 16 bit arithmetic from being rounded to zero before the parameter update.

4.5.5 Validation and Early Stopping

To monitor the model's generalization capabilities and detect potential overfitting, validation is performed every five epochs. This frequency is chosen over per epoch validation because of the high computational cost of the latter, particularly in patch-based mode where sliding window inference must be applied to each validation image individually. At the last epoch validation is always performed regardless of the frequency setting.

During validation no augmentation is applied and no gradient is computed. In patch-based mode, each validation image is processed through sliding window inference as described in Section 4.5.6, producing a full resolution heatmap that is then compared against the ground truth mask to compute the loss and metrics. In resize mode, images are simply rescaled to the training resolution and processed in a single forward pass.

The metrics computed at each validation epoch are: Dice Loss, Dice Score, Precision, Recall, F1 Score, Intersection over Union (IOU), and the Area Under the ROC Curve (AU-ROC). These metrics are described in detail in Section 5.1.3 of Chapter 5.

The checkpointing system maintains two weight files throughout the training process: the checkpoint from the most recently completed epoch and the checkpoint from the best-performing epoch. The criterion for identifying the best epoch depends on the loss normalization strategy. When Exponential Moving Average (EMA) normalization is active, the validation Dice Loss is used as the evaluation metric, since the dynamically rescaled total normalized loss cannot be consistently compared across epochs. Otherwise, the unnormalized total validation loss is used. To avoid saving checkpoints for negligible fluctuations, a new best epoch is recorded only if the improvement exceeds a relative threshold of 0.5% compared to the previous best. Finally, to prevent unnecessary computation when the model has stopped learning, early stopping [17] is triggered, and training is terminated if no improvement is observed for two consecutive validation checks (i.e., a patience of 10 epochs).

4.5.6 Inference Strategy

At inference time the strategy differs depending on whether the model was trained in resize mode or patch-based mode. Figure 4.3 shows a comparison between the two approaches.

In **resize mode**, the test image is rescaled to the same resolution used during training, passed through the model in a single forward pass, and the resulting heatmap is then upsampled back to the original image resolution using bilinear interpolation. The ground truth mask is similarly upsampled to the original resolution before computing evaluation metrics, so that predictions and ground truth are always compared at full resolution.

In **patch-based mode**, the full resolution image is processed using sliding window inference. A window of fixed size (matching the training crop size) is slid across the image with a stride equal to half the window size, producing a set of overlapping crops. For each crop, the model produces a heatmap of the same spatial size as the crop. Where crops overlap, multiple predictions are available for the same pixel. These overlapping predictions are blended using Gaussian weighting, implemented through the MONAI library: pixels near the centre of a crop receive a higher weight than pixels near the edges,

under the assumption that the model is more reliable in the central region of each crop where boundary effects are absent. The final heatmap for the full image is the weighted sum of all crop predictions, at the original image resolution. No resizing of the original image is ever performed in this mode.

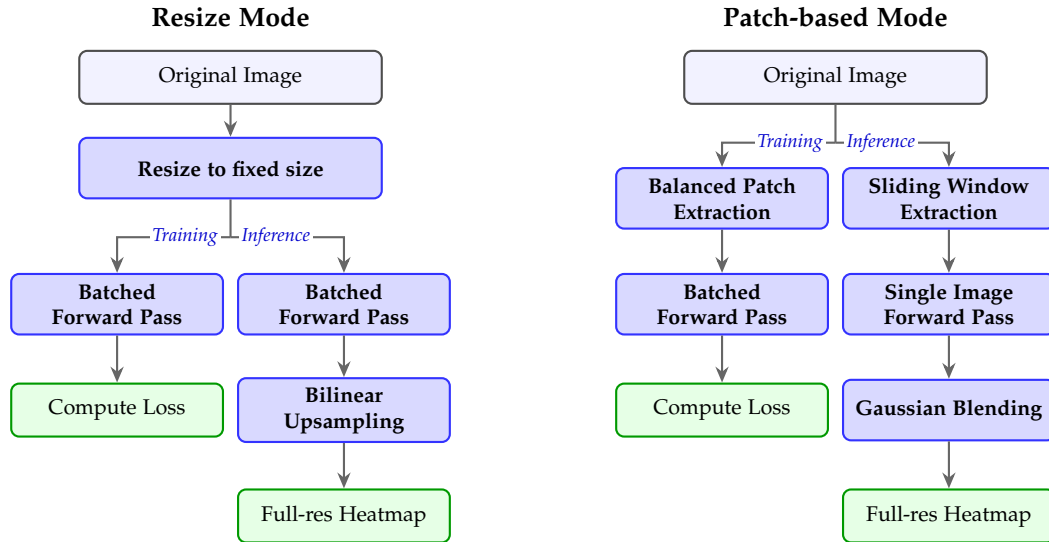


Figure 4.3: Detailed pipeline of the data processing and execution modes. The framework dynamically branches based on the phase (Training vs. Inference) and the chosen input mode (Resize vs. Patch-based), adapting its extraction logic and objective appropriately.

Chapter 5

Experimental Results

In this chapter, we present a comprehensive evaluation of the proposed Dinolizer framework. We begin by assessing the in-distribution localization performance across different backbones, classification heads, and inference modes on the SAGI-D dataset. We then analyze the training dynamics and conduct ablation studies on the objective function and data augmentation pipeline. Subsequently, we evaluate model robustness and generalization capabilities with OOD tests. Finally, we provide a qualitative visual analysis of the model’s predictions across various manipulation scenarios.

5.1 Experimental Setup

5.1.1 Hardware and Accessibility

A primary motivation of this work is to provide an effective inpainting detection tool that does not require prohibitive computational resources. While modern generative models and large-scale detectors often demand massive GPU clusters or expensive cloud instances, our approach is explicitly designed to be trainable and executable on consumer-grade hardware.

All experiments, including the training of the classification heads and the full-resolution sliding window inference, were conducted locally on a single machine equipped with an NVIDIA GeForce RTX 3050 GPU featuring 8 GB of VRAM. To adhere to these memory constraints, the DINOv3 backbone was kept strictly frozen. Furthermore, the entire pipeline was optimized using mixed precision training (`torch.autocast` with 16-bit floating-point arithmetic) and gradient scaling. This dramatically reduced the memory footprint, allowing for practical batch sizes and fast iteration cycles while maintaining similar numerical stability and the localization performance of the network.

To validate the computational accessibility of the proposed pipeline, both training and inference requirements were carefully tracked. During the training phase, the resize-based mode proved highly efficient, requiring approximately 10 to 15 minutes per epoch. In contrast, the patch-based sliding window approach required roughly 1.5 hours per epoch (totaling approximately 24 hours for a full cycle of 20 epochs), as VRAM limitations restricted the maximum batch size to 8 during these high-resolution runs. At inference time, the efficiency of the frozen foundation model paradigm becomes particularly evident. Operating in resize mode at a 640x640 resolution with a batch size of 8, the model achieved an average inference speed of 0.29 seconds per image. When switching to the patch-based inference, the batch size is inherently forced to 1 to accommodate images of variable native dimensions. Despite this constraint, the process remains remarkably efficient, requiring approximately 0.48 seconds per image and completing a test set of over 2200 images in roughly 18 minutes. Most notably, the peak VRAM allocation during both testing phases

Hyperparameter	Value
Training epochs	20
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-3}
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 4)
Augmentation prob.	0.3
Validation frequency	Every 5 epochs
Early stopping patience	2 validation checks (10 epochs)
Batch size	8
Mixed precision	fp16 (torch.amp)
Focal loss α	0.6
Focal loss γ	1.5
EMA momentum	0.98

Table 5.1: Hyperparameter configuration shared across all experiments. Focal loss and EMA parameters apply only to runs using the combined Focal + Dice loss with EMA normalization.

remained strictly contained between 2.2 GB and 2.3 GB, utilizing only a fraction of the available hardware memory.

Finally, the experimental environment demanded significant storage overhead, requiring approximately 100 GB of disk space to accommodate the software dependencies, the foundation model weights, and the large-scale inpainting datasets.

5.1.2 Training Configuration

In our primary experiments, we performed 20 epochs of resize-based training for each of the model variants summarized in Table 4.1. These runs utilized our comprehensive baseline configuration, detailed in Table 5.1, defined as follows: we employed the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-3} to strictly penalize overfitting. A ReduceLROnPlateau scheduler was utilized to halve the learning rate if the validation loss failed to improve for 4 consecutive validation checks. During training, spatial data augmentation (random flips, rotations, color jitter, and Gaussian blur) was applied jointly to the image and the mask with a probability of $p = 0.3$ (each modification had a 0.3 probability of being applied to a given image, and the same image was subjected to different alterations in different epochs).

To prevent unnecessary computation, early stopping was implemented with a patience of 2 validation checks (triggering after 10 epochs without improvement, given a validation frequency of 5 epochs). A batch size of 8 was maintained across both resize and patch-based modes to ensure experimental consistency, although the resize mode could technically accommodate a batch size of 16. Furthermore, 16-bit mixed precision (torch.amp) was utilized to reduce the computational overhead without compromising the network’s localization accuracy. The network was trained using the combined loss function detailed in Section 4.5.3, incorporating EMA normalization and default Focal Loss parameters ($\alpha = 0.6$ and $\gamma = 1.5$).

Following the initial resize-based evaluation, subsequent patch-based training sessions were conducted on the top-performing architectures using an identical hyperparameter setup. Model checkpointing automatically preserved only the weights from the epoch that achieved the lowest validation Dice Loss.

5.1.3 Evaluation Metrics

The evaluation of inpainting localization is performed using pixel-level metrics. The raw probability heatmaps produced by the model (ranging from 0 to 1) are binarized using a strict threshold $\tau = 0.5$. A pixel with a predicted probability ≥ 0.5 is classified as manipulated (Positive), while a pixel < 0.5 is classified as authentic (Negative). Based on this classification, we compute the number of True Positive (TP), False Positive (FP), and False Negative (FN) pixels accumulated globally across the entire test set. The models are then evaluated using the following standard metrics:

- **Precision:** The proportion of predicted inpainted pixels that are actually inpainted.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP} + \varepsilon}.$$

- **Recall:** The proportion of actual inpainted pixels that the model successfully detected.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN} + \varepsilon}.$$

- **F1 Score:** The harmonic mean of Precision and Recall, providing a balanced measure of the model's accuracy.

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall} + \varepsilon}.$$

- **Intersection over Union (IoU):** Also known as the Jaccard Index, it measures the strict overlap between the predicted mask and the ground truth.

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN} + \varepsilon}.$$

- **Dice Score:** Statistically equivalent to the F1 score in binary segmentation, heavily used in localization tasks.

$$\text{Dice} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN} + \varepsilon}.$$

where $\varepsilon = 10^{-8}$ is a small constant to avoid division by zero. All metrics are computed globally, meaning the TP, FP, and FN counts are accumulated across all images in the test set before computing the final ratios. This is equivalent to a micro-averaged evaluation and avoids giving disproportionate weight to images with very small manipulated regions.

Additionally, we report the **Area Under the Receiver Operating Characteristic Curve (AUROC)**. Unlike the previous metrics, AUROC measures the model's ability to rank inpainted pixels above authentic ones at all possible thresholds, operating directly on the continuous probability heatmaps without requiring a predefined threshold. AUROC therefore captures the quality of the continuous probability map rather than only the binary prediction.

5.2 In-Distribution Localization Performance

To evaluate the core localization capabilities of the proposed architectures, we first present the in-distribution results on the test split of the SAGI-D dataset. We compare the models trained in resize mode (processing the full image rescaled to either 320×320 or 640×640) against those employing patch-based sliding window inference. Due to higher time and resource requirements for patch-based training, we only trained a subset of the total models for our evaluation.

5.2.1 Resize-Based Inference

Table 5.2 summarizes the performance of all variants of models trained and evaluated in resize mode. This approach is computationally faster and more scalable than the patch-based alternative, as it provides a global view of the image. However, resizing images to a fixed square dimension inherently distorts aspect ratios and destroys microscopic inpainting traces, making the localization task severely more difficult.

Backbone	Head	Size	Precision	Recall	F1	IoU	Dice	AUROC
DINOv3 ViT-S/16	Linear	320×320	0.6879	0.7741	0.7284	0.5729	0.7284	0.9104
DINOv3 ViT-S/16	MLP	320×320	0.8141	0.7973	0.8056	0.6745	0.8056	0.9480
DINOv3 ViT-S/16	CNN	320×320	0.7718	0.7906	0.7811	0.6408	0.7811	0.9408
DINOv3 ConvNeXt-T	Linear	640×640	0.8112	0.8278	0.8195	0.6941	0.8195	0.9600
DINOv3 ConvNeXt-T	MLP	640×640	0.8051	0.8357	0.8201	0.6951	0.8201	0.9603
DINOv3 ConvNeXt-T	CNN	640×640	0.8117	0.8266	0.8191	0.6936	0.8191	0.9593
DINOv3 ConvNeXt-S	Linear	640×640	0.7836	0.8585	0.8194	0.6940	0.8194	0.9625
DINOv3 ConvNeXt-S	MLP	640×640	0.7813	0.8535	0.8159	0.6890	0.8159	0.9609
DINOv3 ConvNeXt-S	CNN	640×640	0.7858	0.8474	0.8154	0.6884	0.8154	0.9604
DINOv3 ViT-S/16	MLP	640×640	0.8314	0.8445	0.8379	0.7210	0.8379	0.9636

Table 5.2: Quantitative localization results on the SAGI-D test set in resize mode. Metrics are computed globally at the pixel level.

Initially, the transition from the Transformer-based ViT architecture (processed at 320×320) to the CNN-based ConvNeXt architecture (processed at 640×640) appeared to produce a significant performance leap, pushing the overall AUROC from 0.94 to 0.96 and improving all the metrics across the board. We purposely chose these differing resolutions to ensure both models produced a probability estimate grid of the same spatial dimension before upsampling, theoretically aiming for a fair structural comparison. However, this comparison unfairly penalized the ViT model, as the extreme downsampling likely suppressed the fine details necessary for forgery detection.

To establish a genuinely fair comparison, we retrained our best-performing Vision Transformer (DINOv3 ViT-S/16 with an MLP head) at the higher 640×640 resolution to evaluate the architectural differences on identical inputs. The newly trained model showed vast improvements over its lower-resolution baseline, slightly outperforming the ConvNeXt approaches and highlighting the detrimental effect that aggressive downsampling has on inpainting traces. A key architectural difference worth noting is that at a shared input resolution the ConvNeXt models produce a smaller spatial grid of higher-dimensional tokens, whereas the Vision Transformer generates a larger grid of tokens with lower dimensionality. This explains why the ViT relies on the non-linear MLP head to extract meaningful representations.

Indeed, the choice of the classification head affects the backbones differently. For the ViT backbone, the representational capacity of the head is crucial: the MLP architecture significantly outperforms the basic Linear Head, indicating that the non-linearity of the MLP head is highly beneficial for decoding the ViT patch embeddings and to produce accurate manipulation maps. Conversely, for both ConvNeXt variants, the performance across the three heads remains practically identical. Since ConvNeXt intrinsically aggregates local context through its internal convolutional stages, the resulting token embeddings are already highly expressive. Adding complex MLP or CNN heads on top of a ConvNeXt backbone provides no tangible benefit, and the computationally cheaper Linear head is sufficient to reach peak performance.

5.2.2 Patch-Based Inference

The resize approach alters the aspect ratio and spatial dimensions of the images, which could negatively impact the model’s ability to accurately localize inpainted regions. To

address this problem, we trained some selected models using the patch-based approach detailed in Section 4.5. Table 5.3 details the performance of the architectures when evaluated using a 512×512 sliding window, allowing the models to process the images at their native resolution without downsampling artifacts. The training was performed using the same configuration and hyperparameters described in Table 5.1.

Backbone	Head	Size	Precision	Recall	F1	IoU	Dice	AUROC
DINOv3 ViT-S/16	Linear	512×512	0.5941	0.8615	0.7032	0.5423	0.7032	0.9191
DINOv3 ViT-S/16	MLP	512×512	0.7911	0.8812	0.8337	0.7149	0.8337	0.9683
DINOv3 ViT-S/16	CNN	512×512	0.8044	0.8706	0.8362	0.7185	0.8362	0.9672
DINOv3 ConvNeXt-S	Linear	512×512	0.7486	0.9152	0.8236	0.7001	0.8236	0.9724

Table 5.3: Quantitative localization results on the SAGI-D test set in Patch-based mode (sliding window inference). Metrics are computed globally at the full original resolution of the images.

In theory, the patch-based approach should yield significant performance improvements, as the input image resolution is preserved and microscopic inpainting traces remain unaltered. Furthermore, the model processes multiple high-resolution crops per image during training, providing a large number of examples of manipulated boundaries. However, the results in Table 5.3 reveal a different outcome: the improvements are marginal relative to the intense computational cost, particularly for the Transformer-based architectures.

This behavior is consistent with the hypothesis that Vision Transformers could suffer a structural limitation when applied to isolated image crops. ViTs rely on global self-attention to build semantic context, therefore by reducing their field of view to a 512×512 region, they are deprived of the necessary global information to resolve ambiguities. Even though sliding window inference eventually aggregates patch predictions to reconstruct the full spatial grid, the model’s lack of global context during training hinders its ability to detect broad semantic inconsistencies. As evidence consistent with this hypothesis, the MLP variant of the Transformer trained in resize mode at 640×640 resolution (Table 5.2) slightly outperforms its patch-based counterpart in nearly all threshold-dependent metrics, achieving similar results at a fraction of the training cost.

Due to strict computational constraints, patch-based training for the ConvNeXt category was reserved exclusively for the top-performing baseline: the ConvNeXt-Small Linear. Unlike the ViT, the ConvNeXt backbone inherently aggregates local context through deep convolutional layers, making it significantly more robust to the limited field of view of isolated patches. When comparing the patch-based ConvNeXt to its resize-based counterpart, an interesting dynamic emerges within the threshold-dependent metrics. The sliding window approach trades a noticeable drop in Precision (from 0.7836 to 0.7486) for a substantial increase in Recall (from 0.8585 to 0.9152). This indicates that processing images at native resolution in isolated patches makes the model highly sensitive to local manipulation artifacts, significantly reducing false negatives, but at the cost of generating more false positives in authentic regions.

While this heightened sensitivity yields a marginal improvement in the F1 score (0.8236) and achieves the highest peak AUROC (0.9724), an analysis of spatial overlap provides a different perspective. When evaluating the Intersection over Union (IoU) and Dice scores, the resize-based Vision Transformer trained at 640×640 (Table 5.2) still firmly outperforms the patch-based ConvNeXt. This suggests that sliding window inference is a powerful tool capable of increasing recall, but providing the network with the global image context at a sufficiently high resolution remains the most effective strategy for achieving a balanced and highly precise localization of the manipulated regions.

5.3 Extended Training of the Best Architecture

Given the results obtained from the previous experiments (Section 5.2), we identified the DINOv3 ViT-S/16 model with the MLP head trained at 640×640 resolution as the most promising architecture, because it achieved the highest F1, IoU and AUROC among the resize-based configurations, outperforming even the patch-based variants on most metrics, despite its training costs a fraction of the time and resources. We therefore extended its training up to epoch 50 to verify if it would further improve its baseline performance.

5.3.1 Overall Performance and Training Dynamics

Backbone	Head	Epochs	Precision	Recall	F1	IoU	Dice	AUROC
DINOv3 ViT-S/16	MLP	20	0.8314	0.8445	0.8379	0.7210	0.8379	0.9636
DINOv3 ViT-S/16	MLP	50	0.8494	0.8587	0.8541	0.7453	0.8541	0.9701

Table 5.4: Comparison of the DINOv3 ViT-S/16 (MLP head, 640×640 resize mode) performance at 20 and 50 epochs.

Table 5.4 shows the comparison between epoch 20 and epoch 50 for our chosen architecture. The additional 30 epochs of training allowed the network to refine its boundary detection, improving both Precision and Recall. To thoroughly understand this learning behavior, we tracked the evolution of the objective functions throughout the optimization process. Figure 5.1 illustrates the training and validation trajectories for both the Dice Loss and the Focal Loss over the 50-epoch run.

Analyzing the Dice Loss, we observe a classic convergence pattern where the training loss decreases steadily. However, a noticeable generalization gap emerges between the training and validation curves. This indicates a mild degree of overfitting, which validates our rigorous regularization strategy. To actively combat this problem, we employed an aggressive spatial data augmentation, combined with a Dropout of $p = 0.3$ and AdamW weight decay; without these constraints, the network would likely have memorized the training distribution, resulting in a significantly wider divergence.

The Focal Loss dynamics exhibit a contrasting behavior, where the validation loss starts and remains consistently lower than the training loss. During training, our data augmentation pipeline (specifically random Gaussian blur and color jitter) artificially obscures manipulation boundaries, increasing the number of "hard" pixels and inflating the Focal Loss. Conversely, validation is performed on pristine images where boundaries remain sharper, resulting in a naturally lower and more stable trajectory.

Finally, we can observe how the two losses intrinsically operate at different scales: the Dice Loss fluctuates around 0.5, whereas the Focal Loss operates on a considerably smaller scale, around 0.05. This order-of-magnitude discrepancy empirically validates our choice to employ Exponential Moving Average (EMA) normalization, without which the gradients produced by the Focal Loss would be strongly dominated by the Dice Loss ones. As observed in Figure 5.1, both loss values steadily decrease over the extended training process, supporting the stability of the optimization.

5.3.2 Robustness Analysis

Our portion of the SAGI-D dataset, described in Section 3.5, contains images inpainted using five different state-of-the-art architectures. To ensure the model has learned generalized forgery features rather than overfitting to the artifacts of a specific generative architecture, we decomposed the test metrics across these five distinct tools, evaluating the results obtained from our best model, the one with the lowest validation loss during a 50-epoch training.

Notably, the splitting process employed for the creation of the dataset is stratified around the image generator. The test dataset comprises a total of 1480 images, evenly distributed among the generators (300 images each), with the exception of HDPainter (280 images). Table 5.5 reports the localization performance across generators, highlighting a varied landscape of detection difficulty. The network achieves peak accuracy against *RemoveAnything* (F1 = 0.9236, AUROC = 0.9913), suggesting this tool leaves highly discernible artifacts. Conversely, *InpaintAnything* emerges as the most sophisticated adversary, with an F1 score of 0.6953. Despite these expected algorithmic variations, the model maintains a strong baseline detection capability across all five distinct generators, supporting its broad applicability.

To contextualize these quantitative metrics, Figure 5.2 showcases the predictions of the model across the different inpainting tools. The visual representation only shows one image per generator and therefore is not a statistically accurate representation of the general model’s capability against a specific inpainting algorithm. However, it can be useful to provide a general idea of its performance. For the generator that yielded the highest detection performance, the progression reveals highly precise masks with strongly polarized heatmaps. Even in the more challenging scenarios generated by *InpaintAnything*, the network successfully localizes the core of the manipulation, though the predicted boundaries exhibit a higher degree of uncertainty.

Generator	Images	Precision	Recall	F1	IoU	Dice	AUROC
GLOBAL	1480	0.8494	0.8587	0.8541	0.7453	0.8541	0.9701
BrushNet	300	0.7977	0.8186	0.8080	0.6779	0.8080	0.9618
HDPainter	280	0.8213	0.7765	0.7983	0.6643	0.7983	0.9510
PowerPaint	300	0.8503	0.8679	0.8590	0.7529	0.8590	0.9703
InpaintAnything	300	0.7440	0.6526	0.6953	0.5329	0.6953	0.9192
RemoveAnything	300	0.8919	0.9576	0.9236	0.8580	0.9236	0.9913

Table 5.5: Localization metrics of the best model (ViT-Small MLP, 50 epochs) decomposed by generative model, including the test split distribution.

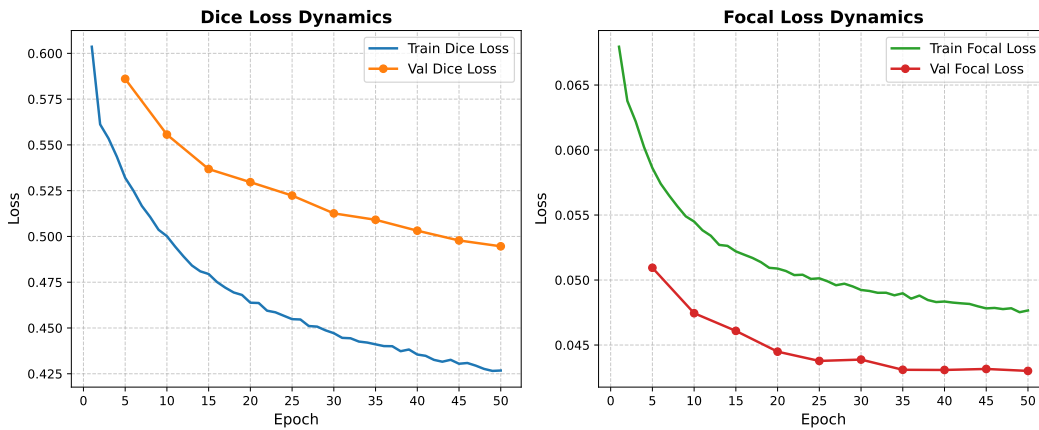


Figure 5.1: Training and validation loss trajectories over 50 epochs. The Dice Loss is shown on the left and the Focal Loss on the right.

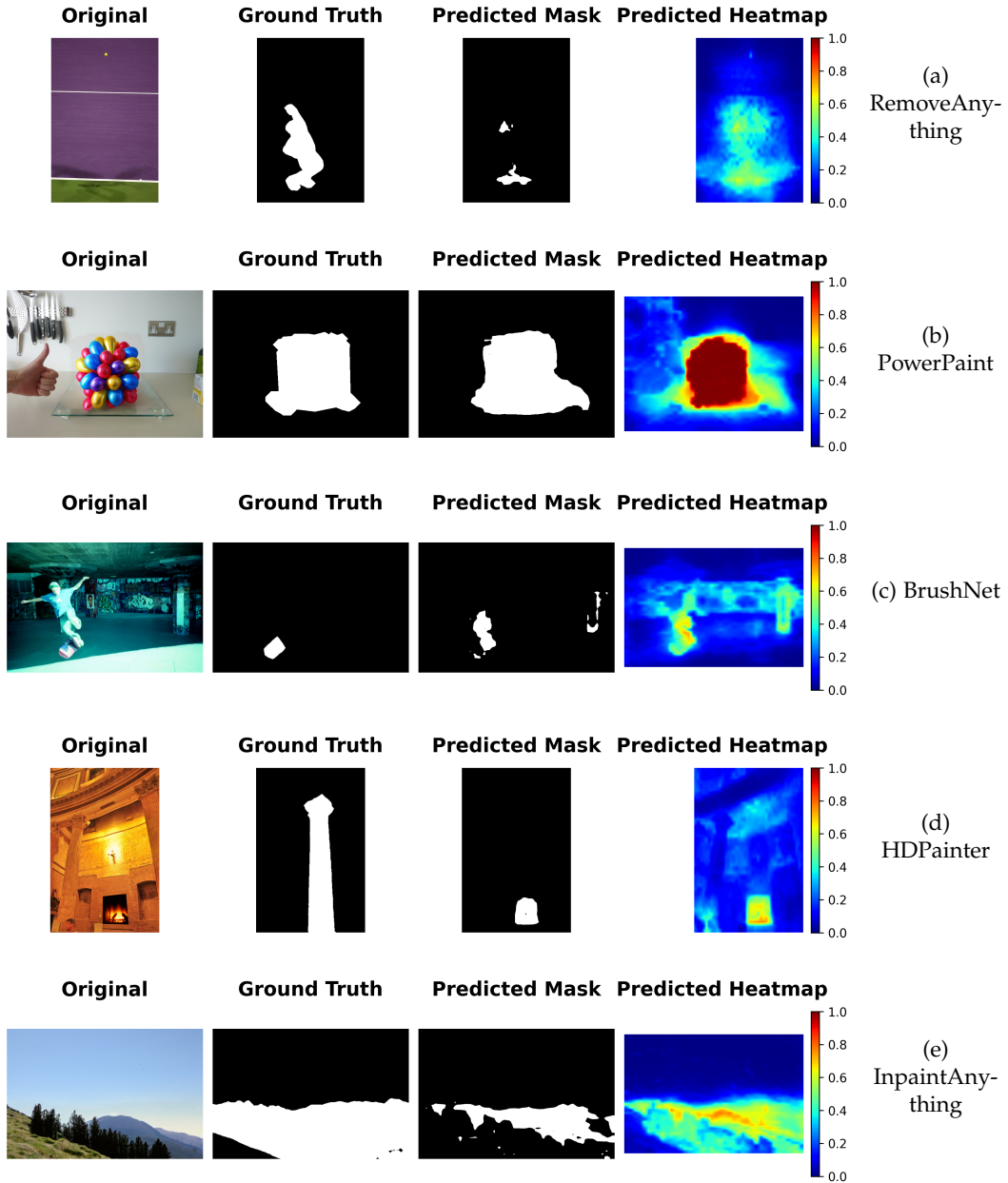


Figure 5.2: Qualitative localization results of the ViT-Small MLP model across the five distinct generative models present in the SAGI-D test set. The columns represent (from left to right): Input Image, Ground Truth Mask, Predicted Binary Mask, and Predicted Probability Heatmap.

5.4 Ablation Studies

To empirically validate the structural and algorithmic choices of the Dinolizer framework, we conducted several ablation experiments. By removing isolated components from the comprehensive baseline described in Table 5.1, we were able to quantify the individual impact of each strategy on the final performance and training stability.

All ablation tests were performed using the ConvNeXt-Small backbone with a Linear classification head in resize mode (640×640), which represented our most efficient and best-performing baseline at the time of the experiments. While these ablations are focused on the resize-based approach due to computational constraints, the structural dynamics observed are expected to translate consistently across different backbones and patch-based inference strategies. Table 5.6 contains a comparison of the performance of the comprehensive baseline against the three distinct ablation configurations designed in this study:

1. **Dice + Focal Loss (Raw Sum):** The model utilizes both loss functions, but the Exponential Moving Average (EMA) normalization is deactivated.
2. **Dice Loss Only:** The model is trained relying exclusively on the Dice Loss, entirely removing the pixel-level classification focus provided by the Focal Loss.
3. **No Augmentation:** The joint spatial transformations (color jitter, blur, and geometric rotations) are disabled, exposing the model to the unmodified training dataset.

Configuration	Precision	Recall	F1	IoU	Dice	AUROC
Comprehensive Baseline	0.7836	0.8585	0.8194	0.6940	0.8194	0.9625
Dice + Focal Loss (Raw Sum)	0.7696	0.8671	0.8154	0.6884	0.8154	0.9605
Dice Loss Only	0.7852	0.8427	0.8130	0.6849	0.8130	0.9365
No Augmentation	0.7813	0.8654	0.8212	0.6966	0.8212	0.9632

Table 5.6: Ablation study on the objective function and data augmentation.

5.4.1 Objective Function Analysis

Evaluating the objective function components, the configuration utilizing the raw sum of Dice and Focal Loss (without EMA normalization) highlights a severe optimization imbalance. As observed in the training dynamics (Figure 5.3, middle row), the unnormalized Dice Loss operates on a significantly larger scale compared to the unnormalized Focal Loss. Consequently, the gradient updates are almost exclusively driven by the Dice component. This scale discrepancy causes the validation Focal Loss to stagnate and slightly worsen as training progresses, indicating that without dynamic EMA scaling, the multi-loss strategy loses its regularizing effect.

Removing the Focal Loss entirely (Dice Loss Only) results in a noticeable drop in AUROC (from 0.9625 to 0.9365). This indicates that the Focal Loss is beneficial for improving the pixel-level continuous probability estimation, effectively penalizing the hard-to-classify pixels at the manipulation boundaries. However, evaluating the overall quantitative results in Table 5.6, it is important to note that while the introduction of the Focal Loss and EMA normalization provides measurable benefits in terms of training stability and AUROC, their impact on the threshold-dependent metrics (such as the F1 score) remains marginal. The comprehensive multi-loss strategy ensures a more balanced optimization process, but does not drastically alter the fundamental detection capabilities of the backbone. Future work should explore comprehensive hyperparameter tuning (e.g., adjusting the focal α and γ parameters) to definitively assess whether the inclusion of focal loss justifies the added optimization overhead.

5.4.2 Data Augmentation Impact

The ablation on data augmentation presents a seemingly counterintuitive result: the *No Augmentation* model achieves slightly higher in-distribution metrics (F1 score of 0.8212) compared to the baseline’s 0.8195. However, a visual analysis of the training dynamics (Figure 5.3, bottom row) reveals the underlying problem.

Without augmentation, the validation loss curves detach from the training trajectories almost immediately (around epoch 5), indicating rapid and severe overfitting. The network quickly memorizes the pristine pixel distributions and generator-specific noise patterns of the SAGI-D dataset. While this memorization inflates the performance on the in-domain validation and test sets, it fundamentally undermines the model’s ability to learn robust, generalized forensic features. For this reason, data augmentation remains necessary to reduce overfitting and improve robustness, even when it does not directly benefit the in-distribution test metrics.

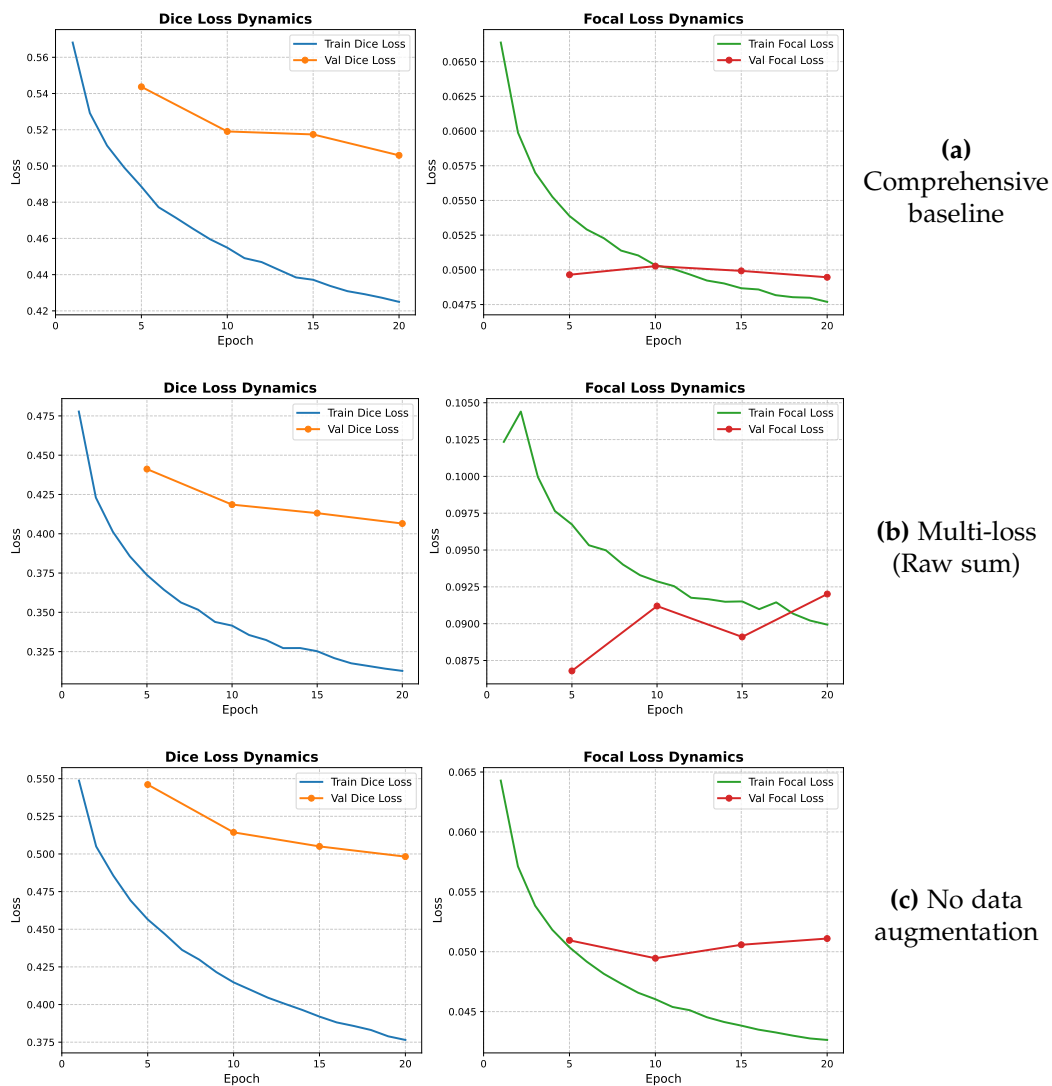


Figure 5.3: Training dynamics comparison: **(a)** Comprehensive baseline with stable convergence; **(b)** Raw Sum multi-loss exhibiting stagnation in the validation Focal Loss; **(c)** No Augmentation showing immediate and severe overfitting.

5.5 Out-of-Distribution Testing

To rigorously evaluate the generalization capabilities of our architecture, we conducted Out-of-Distribution (OOD) testing. Deep learning models applied to image forensics often suffer from severe dataset bias, learning to detect the specific artifacts of the training generators rather than a universal concept of image forgery.

To quantify this phenomenon, we measured the performance of our best ViT and ConvNeXt models trained exclusively on the SAGI-D dataset (Section 3.5) against the two additional datasets introduced in Chapter 3: the comprehensive Flickr30k dataset (Section 3.4), and our custom BtB-COCO dataset (Section 3.3). Crucially, both OOD datasets were generated using the *Beyond the Brush* pipeline powered by Fooocus (a state-of-the-art Stable Diffusion XL architecture). This allows us to test the models’ robustness against a highly sophisticated generative engine that was entirely absent from the training set, observing their behavior across two distinct source image domains.

Additionally, we verified whether the ConvNeXt architecture trained without data augmentation (as explored in Section 5.4) would exhibit further degradation when evaluated on external datasets. Finally, we trained our optimal architecture on a combined domain comprising both SAGI-D and Flickr30k. We then evaluated its performance on a unified test set to assess how the presence of multiple domains affects the performance on the individual generators.

5.5.1 ViT Backbone Evaluation

Dataset	Precision	Recall	F1	IoU	Dice	AUROC
SAGI-D (In-Domain)	0.8494	0.8587	0.8541	0.7453	0.8541	0.9701
Flickr30k (OOD)	0.4509	0.1857	0.2631	0.1515	0.2631	0.7024
COCO Fooocus (OOD)	0.4948	0.2949	0.3696	0.2267	0.3696	0.7864

Table 5.7: Quantitative comparison highlighting the severe performance degradation when evaluating the model on Out-of-Distribution datasets compared to the in-domain test.

As detailed in Table 5.7, the performance drop across both external datasets is severe. On the Flickr30k set, the F1 score collapses to 0.2631, with Recall plummeting to 0.1857. The network becomes extremely conservative, missing the vast majority of the manipulated pixels. The performance on the BtB-COCO dataset shows a slight improvement (F1 score of 0.3696), but the overall trend of inaccurate predictions remains consistent. This alignment is expected, as both datasets were generated using the same underlying state-of-the-art generative engine. The network struggles to identify manipulations produced by Fooocus, likely due to the smoother boundaries and superior semantic coherence between the inpainted regions and the rest of the image, characteristic of SDXL. The minor variance between the two datasets could be attributed to the updated prompt generation model in our modified BtB pipeline, which may suggest slightly less contextually coherent replacements, inadvertently facilitating the Vision Transformer’s localization task.

The qualitative analysis in Figures 5.4 and 5.5 provides crucial insights into the network’s failure modes. The predicted heatmaps indicate that the model is not predicting random noise; rather, it often correctly flags isolated boundary anomalies, such as the artificial transition edges where the generated pixels meet the authentic background, or microscopic texture inconsistencies like unnatural pixel grain. However, the model is often not confident enough to correctly identify the inner core of the manipulation. This suggests that the network may have learned specific boundary and frequency artifacts created by the generators used in our SAGI-D model. These features might be absent or distributed differently in the Fooocus generator, causing low generalization across inpainting methods.

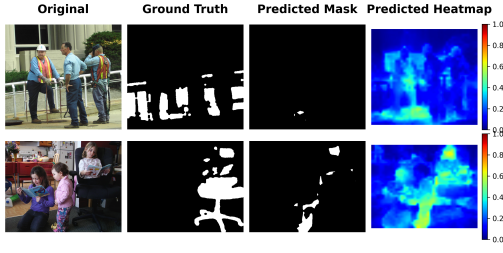


Figure 5.4: Failure cases on the Flickr30k dataset. The model detects faint boundary anomalies but outputs nearly blank masks.

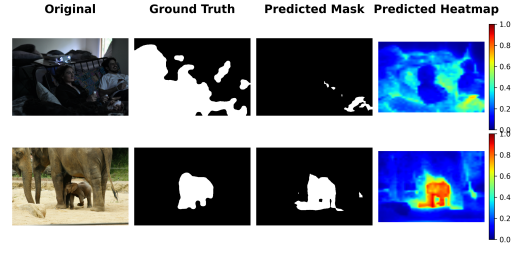


Figure 5.5: Predictions on the COCO Foocus dataset. The network highlights manipulated shapes but lacks confidence.

5.5.2 ConvNeXt OOD Evaluation and Impact of Data Augmentation

To further investigate this vulnerability and evaluate the impact of our regularization strategies, Table 5.8 details the performance of the ConvNeXt-Small baseline compared to its non-augmented ablation counterpart when exposed to the same Foocus-generated datasets.

Much like the ViT architecture, the comprehensive ConvNeXt baseline suffers a substantial performance drop when shifting from the in-domain SAGI-D test set to the OOD datasets. A common assumption in deep learning is that spatial and color transformations force the network to learn generalized features, inherently improving cross-domain robustness. However, testing the *No Augmentation* ablated model on the same OOD datasets yields a counterintuitive result: the non-augmented model performs marginally better on external datasets (e.g., F1 of 0.3052 on Flickr30k compared to the baseline’s 0.2852).

This phenomenon highlights a critical limitation in standard regularization techniques applied to image forensics. While data augmentation successfully prevents the memorization of specific noise inside the training domain, adding in-domain regularization does not transfer to out-of-domain robustness. When the source domains are fundamentally disjoint, standard augmentation is insufficient to teach the network the latent artifacts characteristic of unknown, more advanced generative engines, suggesting that true generalization requires exposure to a diverse set of generative architectures during training.

Configuration	Dataset	Precision	Recall	F1	IoU	Dice	AUROC
Comprehensive Baseline	SAGI-D (ID)	0.7836	0.8585	0.8194	0.6940	0.8194	0.9625
	Flickr30k (OOD)	0.4524	0.2082	0.2852	0.1663	0.2852	0.7043
	BtB-COCO (OOD)	0.5028	0.3377	0.4040	0.2531	0.4040	0.7879
No Augment.	SAGI-D (ID)	0.7813	0.8654	0.8212	0.6966	0.8212	0.9632
	Flickr30k (OOD)	0.4427	0.2329	0.3052	0.1801	0.3052	0.7022
	BtB-COCO (OOD)	0.4724	0.3503	0.4023	0.2518	0.4023	0.7876

Table 5.8: Out-of-Distribution evaluation of the ConvNeXt-Small backbone with and without data augmentation. Metrics are reported for the SAGI-D in-domain (ID) test set and the two external OOD datasets.

5.5.3 Multi-Domain Training

Having established that in-domain regularization alone does not guarantee robustness to changes in the generative process and source-image domain, we then investigated whether increasing the diversity of the training data could improve performance across heterogeneous distributions. We therefore constructed a joint training setup using SAGI-D and Flickr30k. The corresponding training splits were merged to form a combined training set, while the validation and test splits were merged separately, preserving the original

train/validation/test separation. This setup increases the diversity of both the source-image content and the inpainting distributions available during training without introducing overlap between the different dataset splits.

We trained our best-performing architecture, the ViT backbone with an MLP classification head operating at a 640×640 resize resolution, on this joint dataset for 50 epochs. The same comprehensive training configuration defined in Table 5.1 was retained, including the EMA-normalized combination of Dice and Focal Loss. We then evaluated the model on the combined test set, extracting the metrics for each individual generative engine to observe how the network scales when exposed to heterogeneous artifacts.

The results detailed in Table 5.9 show that joint training on SAGI-D and Flickr30k substantially changes the model’s performance across the evaluated generative distributions. The global F1 score across the mixed dataset reaches 0.8328, indicating that the ViT backbone is highly capable of simultaneously learning multiple forgery distributions. The most notable observation emerges from the evaluation of the Fooocus (SDXL) images. In the isolated OOD baseline tests, the ViT model completely failed to identify Fooocus manipulations, resulting in a poor F1 score of 0.2631 and Recall of 0.1857. By exposing the network to SDXL artifacts during training, the F1 score increased to 0.6120, and Recall drastically improved to 0.6047. This result indicates that the frozen DINOv3 representations contain sufficient information to support the detection of Fooocus-related artifacts when examples from this domain are included during training.

Despite this significant improvement, Fooocus remains the lowest-performing generator in the combined test set, falling significantly behind models like RemoveAnything (F1 of 0.9238) or PowerPaint (F1 of 0.8338). This performance gap suggests that Fooocus produces manipulations that are more difficult for the proposed detector to distinguish from authentic content, this may be related to the smoother transitions and stronger semantic coherence between its generated regions and the original image.

Generator / Split	Precision	Recall	F1	IoU	Dice	AUROC
Global (Combined Test Set)	0.8355	0.8301	0.8328	0.7134	0.8328	0.9632
RemoveAnything	0.8992	0.9497	0.9238	0.8584	0.9238	0.9902
PowerPaint	0.8343	0.8333	0.8338	0.7150	0.8338	0.9609
HD-Painter	0.7978	0.7583	0.7775	0.6360	0.7775	0.9430
BrushNet	0.7924	0.7534	0.7724	0.6292	0.7724	0.9460
InpaintAnything	0.7311	0.6307	0.6772	0.5119	0.6772	0.9147
Fooocus (SDXL)	0.6194	0.6047	0.6120	0.4409	0.6120	0.8731

Table 5.9: Performance breakdown by generator on the combined test set (SAGI-D + Flickr30k). The model successfully generalizes across multiple domains, though significant variance in detection difficulty remains between generative engines.

Chapter 6

Conclusion

This thesis set out to answer a specific question, posed in Chapter 1: whether pretrained Vision Foundation Models, used strictly as frozen feature extractors, already contain sufficient information to localize generative inpainting, and whether this holds even for comparatively small backbones. The DinoLizer framework proposed in Chapter 4 and evaluated in Chapter 5 answers this question affirmatively. Through a comprehensive experimental pipeline encompassing multiple architectural configurations, distinct inference modes, ablation studies, and multi-domain training, the optimal configuration reached an F1 score of 0.854, an Intersection over Union of 0.745, and an AUROC of 0.970 on the in-distribution SAGI-D test set, while never updating a single parameter of the underlying foundation model.

Four further contributions extend this core result. First, the systematic comparison across head architectures showed that, for the ViT backbone, a non-linear MLP head meaningfully outperforms a simple linear probe, while for the ConvNeXt backbones the three heads perform almost identically. This indicates that the required capacity of the classification head depends directly on how much local context the backbone itself internally aggregates.

Second, the evaluation of different inference strategies highlighted a critical structural trade-off. The resize-based approach provides the network with global semantic context but distorts the aspect ratio and microscopic manipulation traces, whereas the patch-based sliding window inference preserves the native resolution and artifact integrity at the cost of global visibility. The experiments demonstrated that the convolutional nature of ConvNeXt adapts better to the limited field of view of isolated patches, while Vision Transformers heavily rely on the global context provided by the full resized image to accurately resolve semantic ambiguities.

Third, the extensive out-of-distribution evaluation in Chapter 5.5 revealed both the severity and a partial remedy for the generalization gap that affects nearly all inpainting detectors reviewed in Chapter 2. While performance collapses when tested on a generative engine absent from the training set, exposing the model to a highly diverse, multi-domain training environment recovers a substantial part of this gap, more than doubling the F1 score on the previously unseen engine.

Fourth, this entire investigation, from dataset curation to the full ablation and out-of-distribution study, was carried out on a single consumer-grade GPU. This directly demonstrates the accessibility goal stated in Chapter 1, suggesting that the foundation-model forensic approach is not only a viable solution for obtaining significant detection performances, but also a computationally accessible and easily adaptable one.

6.1 Key Results and Their Significance

The central finding of this thesis is that a frozen general-purpose foundation model already encodes strong inpainting-relevant signals. This finding is not unique to this work and is independently corroborated by concurrent work on DinoLizer [14], a DINOv2-based linear-probe detector published during the course of this thesis. The two works were developed independently and evaluated on different datasets and generative engines; therefore, they are not directly comparable on a metric-by-metric basis, and the comparison that follows is deliberately qualitative rather than numerical.

The agreement between the two works is a meaningful result in itself. Both independently arrive at the same core architectural recipe consisting of a frozen self-supervised Vision Transformer decoded by an extremely lightweight classification head. Both report that this configuration is competitive with considerably heavier purpose-built detectors. This convergence, reached from two different datasets and two different backbone generations, strengthens the case that the underlying finding is a genuine property of self-supervised visual representations.

At the same time the two works differ in ways that clarify what makes this thesis a meaningful contribution. DinoLizer builds on a DINOv2 backbone that was already fine-tuned for synthetic image detection on the B-Free dataset prior to being applied to inpainting localization [14]. Furthermore, the DinoLizer authors tested both a linear head and an MLP head but found only marginal improvements with the latter. This thesis instead deliberately starts from a fully general-purpose DINOv3 checkpoint never adapted to any forensic task. Moreover, this thesis introduces and compares a fundamentally different backbone family through the inclusion of the tiny and small ConvNeXt variants, and systematically compares linear, MLP, and convolutional heads across the two backbone families, isolating how much of the localization capability comes from the frozen representation itself versus from task-specific pretraining or head design. As demonstrated in Chapter 5 the convolutional nature of ConvNeXt inherently aggregates local context and renders complex MLP classification heads redundant. This contrasts with the ViT architecture, which benefits from non-linear heads to effectively decode its patch embeddings. The two backbones also react completely differently to the chosen inference mode. ConvNeXt adapts much better to the limited field of view of patch-based sliding window inference. Conversely, the Vision Transformer demonstrates a structural reliance on the global semantic context provided by the fully resized image to accurately resolve manipulations.

The second distinction lies in the scale of the evaluated models and the specific choice of the backbone generation. The DinoLizer authors report their primary results using the DINOv2-Base architecture comprising approximately 86 million parameters. Furthermore, they note in their own ablation study that the older DINOv2 backbone outperforms the newer DINOv3 when training exclusively the classification head. Although this observation provides an interesting insight for future research regarding backbone selection, this thesis deliberately adopts DINOv3 to investigate a different set of architectural properties. Specifically, DINOv3 provides officially supported lightweight architectures such as the ViT-Small with 21.6 million parameters, and most importantly, DINOv3 natively includes the ConvNeXt family, which is entirely absent from the DINOv2 releases. Access to these models was essential to conduct the cross-architecture comparison between Vision Transformers and Convolutional networks that forms a core contribution of this work. Despite utilizing these significantly smaller networks, our approach achieves strong localization performance. This demonstrates that the foundation-model forensic paradigm is not only a viable solution but also highly adaptable to strict computational constraints.

Finally, this thesis incorporates diverse ablation studies, a joint domain training experiment, and extensive cross domain evaluations. The DinoLizer authors evaluate their model across various external datasets, their empirical results confirm a noticeable performance degradation when testing on unseen generators [14]. This thesis observes a similarly severe generalization gap, but takes an additional step by explicitly testing a multi-domain training strategy. The results demonstrate that expanding the training distribution dras-

tically improves localization, specifically on the artifacts introduced by the newly added domain.

Rather than claiming universal generalization across any future architecture, these findings highlight that frozen representations possess the structural capacity to learn and retain multiple distinct generative signatures simultaneously. This positions the systematic study of dataset diversity not as a definitive solution to the out-of-distribution problem, but as a critical and highly effective tool to progressively expand the detection capabilities of the model.

6.2 Failure Cases and Qualitative Analysis

To complement the quantitative metrics of Chapter 5, this section examines the qualitative predictions produced by the multi-domain model at epoch 50. The evaluation was performed on the combined SAGI-D and Fooocus test split. Approximately 300 comparison images were generated and manually inspected to identify recurring patterns of success and failure. Each comparison highlights the original image, the ground truth mask, the predicted binary mask, and the raw probability heatmap.

This qualitative impression aligns closely with the quantitative per-generator breakdown reported in Chapter 5. Fooocus and InpaintAnything are the two lowest performing generators in the evaluation and are disproportionately represented among the weakest predictions inspected in this analysis. RemoveAnything is, on the contrary, the strongest generator quantitatively and one of the most reliably localized in this qualitative inspection. A plausible explanation is that the inpainted regions generated by RemoveAnything frequently retain a faint residual blur around the edited boundary. This provides the model with an additional low level cue alongside the semantic incoherence exploited elsewhere in this analysis.

6.2.1 Strengths: Semantically Coherent Localization

The clearest strength of the proposed approach emerges on images where the manipulated region coincides with a semantically coherent object or a well defined boundary between two distinct visual regions. In these cases the model produces sharp and confident heatmaps that closely track the true boundary.

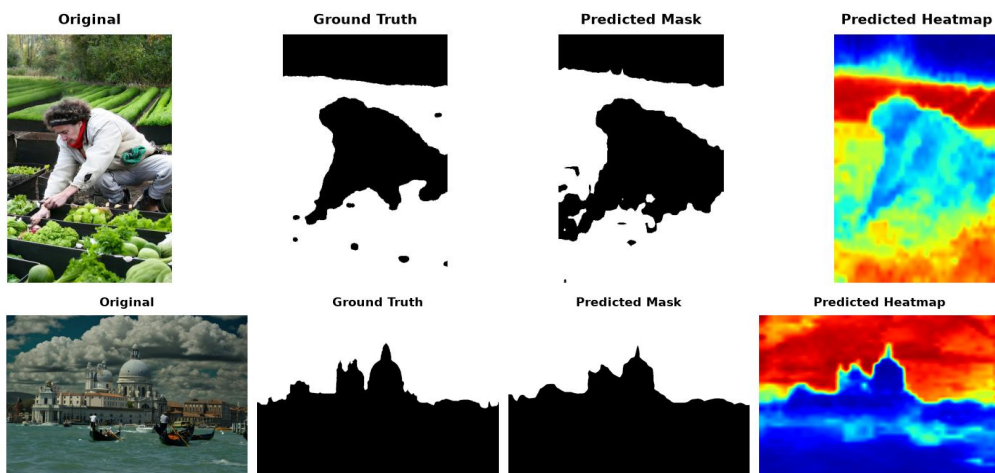


Figure 6.1: Representative successful predictions. Top: a manipulated background is precisely localized while avoiding the unmodified person. Bottom: a massive manipulated sky region is recovered almost perfectly by exploiting the sharp semantic boundary.

This holds true even when the manipulated region is very large or visually camouflaged against its surroundings (Figure 6.1). This behavior is consistent with the hypothesis that the semantic representations learned by the frozen backbone contribute significantly to the localization process.

6.2.2 Failure Mode 1: Confusion Between Manipulation and Visual Saliency

A prominent failure mode observed during the qualitative analysis is the tendency of the network to flag visually salient but entirely authentic objects as manipulated. In these instances the model assigns high probability to prominent items while missing the true manipulated region entirely (Figure 6.2). This pattern emerges in images containing objects that stand out from their surroundings through color saturation or sharp contrast regardless of their authenticity. This behavior indicates that the representation of anomalies that the classification head has learned to associate with manipulation is occasionally entangled with a broader notion of visual saliency. While the model correctly calibrates its confidence in many scenarios to suppress false positives, it remains susceptible to these strong visual distractors. This vulnerability points to a specific and addressable limitation of relying exclusively on high level semantic features without explicit verification of the underlying noise residuals.

6.2.3 Failure Mode 2: Small, Fragmented, or Off-Center Ground-Truth Regions

A second related failure mode occurs on images containing ground-truth masks that are either exceedingly small or composed of disconnected fragments (Figure 6.3). In these cases the predicted mask frequently bears little spatial resemblance to the ground truth. The output appears instead as scattered noise loosely correlated with locally salient content. This behavior could be explained by the coarse spatial resolution at which the classification head operates: since predictions are made on a downsampled grid of patch embeddings before being upsampled to the original resolution, small or thin structures are easily absorbed by neighboring patches.

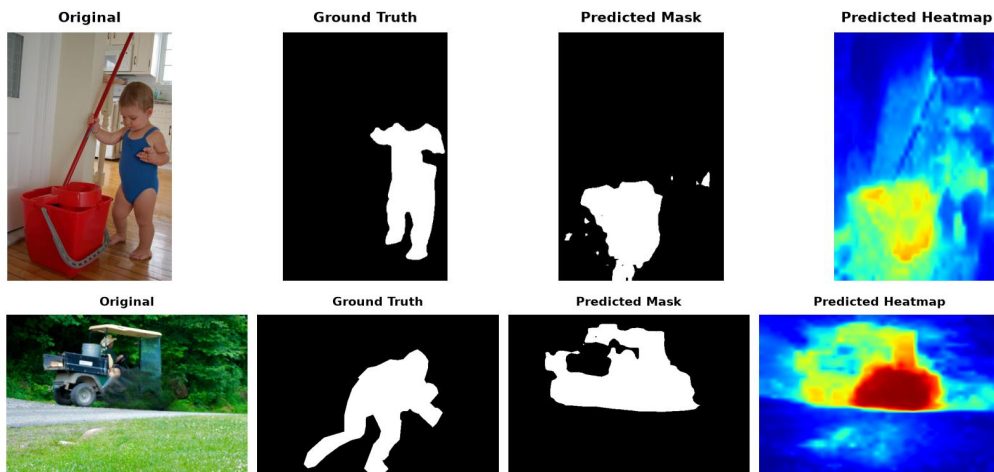


Figure 6.2: Representative failure cases driven by visual saliency. Top: the model assigns higher probability to the bright red bucket while, ignoring the actual manipulation on the child. Bottom: the network incorrectly flags the golf cart, only partially acknowledging the true manipulated grass region.

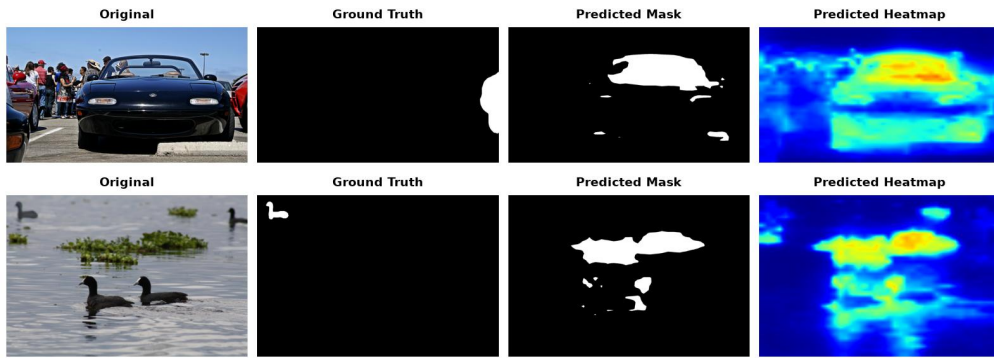


Figure 6.3: Representative failure cases involving small or off-center ground truth masks. Top: a minor edge manipulation is completely missed in favor of the salient car in the foreground. Bottom: a small manipulation on the top left is missed, and the original central grass area is flagged instead.

Furthermore, manipulated regions positioned at the extreme edges of the frame are disproportionately represented among the weakest predictions. A plausible hypothesis is that the model implicitly associates manipulation with regions occupying the visual center of the image and becomes correspondingly less sensitive to incoherence introduced at the periphery. This analysis cannot definitively isolate whether this central bias originates from the representations of the frozen backbone or from the lightweight classification head trained on top of them.

However, it is worth noting that these are tendencies rather than strict rules: several of the best predictions observed in this analysis also involve comparatively small masks, indicating that size alone is not the determining factor, but spatial position and semantic clarity of the manipulated region play an equally important role.

6.2.4 Model Confidence and Boundary Precision

Beyond outright misclassifications, a recurring pattern across the inspected heatmaps is a visible hesitation around the correct region (Figure 6.4): the model frequently assigns moderately elevated probabilities to the true manipulated area, but without surpassing the binarization threshold required to flag it as manipulated in the predicted mask. This suggests that, at least for some manipulation types, the model has partially learned the correct signal but has not yet learned to flag it definitively. This observation suggests that the training signal might have been insufficient, whether in the form of additional epochs, a larger or more diverse training set, or a different augmentation strategy, to force a polarized decision in some ambiguous cases. This directly motivates the future work directions discussed in Section 6.3 regarding data augmentation and alternative loss paradigms.

A second, more structural observation concerns boundary precision. Even in the most successful predictions, the recovered mask boundary is visibly coarser than the ground truth at the pixel level, a direct consequence of the fixed, content-independent patch grid used to tokenize the input. While this coarseness rarely affects whether a manipulated region is identified at all, it does limit the pixel-level precision achievable with the current architecture, and reinforces content-aware tokenization as a promising direction for future refinement rather than a purely theoretical concern.

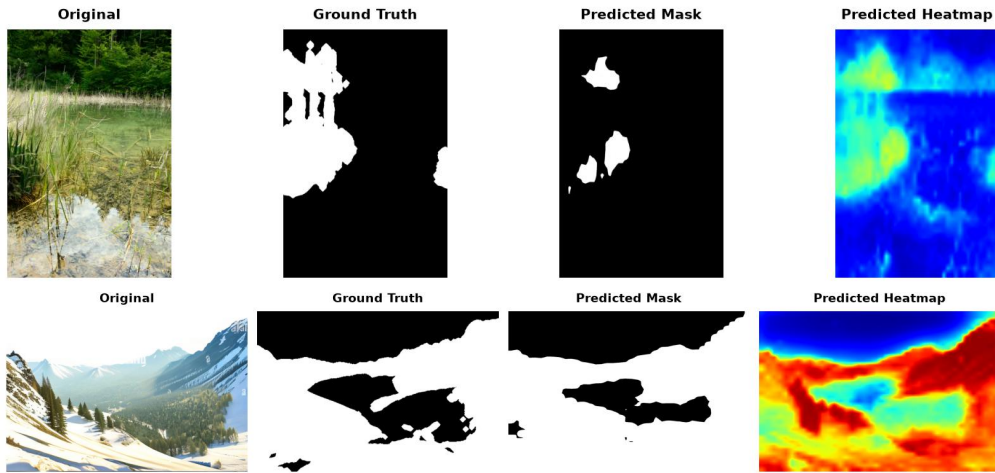


Figure 6.4: Analysis of model confidence and boundary precision. Top: the heatmap correctly highlights the manipulated vegetation, but fails to cross the binarization threshold resulting in a partially correct prediction. Bottom: a successful detection where the predicted boundary exhibits noticeable coarseness and smoothing due to the fixed tokenization grid.

6.3 Limitations and Future Work

While the results presented in this thesis are encouraging, they were obtained under deliberate and significant constraints, and several of these constraints directly suggest concrete directions for future work.

The most immediate constraint is computational. Every experiment in this thesis was conducted on a single consumer-grade GPU with 8 GB of VRAM, a decision that was central to the accessibility goal of this work, but that also bounded the space of configurations that could realistically be explored. A natural extension is therefore to repeat the same systematic comparison with larger backbone variants, such as DINOv3 ViT-B or ViT-L, or larger ConvNeXt configurations, to establish whether the trends observed at the Small scale in Chapter 5 continue to hold, or whether a different balance between backbone capacity and head capacity emerges at larger scale, a question also raised directly by the DINOv2/DINOv3 comparison discussed in Section 6.1.

A second direction concerns the optimization of the training pipeline and its hyperparameters. The comparison in Chapter 4 was limited to the classification head architectures, the loss function configuration, and the data augmentation pipeline used during training; a systematic study of the binarization threshold applied at inference time, together with a broader search over augmentation strategies and different loss paradigms and hyperparameters, could further improve the localization quality without any architectural change.

A third direction concerns the information available to the model. This thesis deliberately isolates the contribution of the frozen backbone’s semantic representation alone, without any dedicated branch for low-level or frequency-domain signals. As discussed in Chapter 2, these are common components of state-of-the-art multi-modal detectors such as TruFor. Training an auxiliary branch in parallel to capture complementary signal, for instance from the frequency domain, and fusing it with the frozen backbone’s representation, is a promising way to test whether the two types of signal are genuinely complementary for inpainting specifically, rather than redundant.

A fourth direction is architectural, and follows directly from the discussion of content-aware region modelling in Chapter 2. The DINOv3 backbones used throughout this work tokenize the input image into a fixed, content-independent square grid; replacing this fixed tokenization with the dynamic, content-coherent region partitioning proposed by

HRGR [46] is a concrete architectural extension that was identified but not implemented in this thesis.

A fifth direction follows from the very recent DINOv3-based baseline discussed in Chapter 2, which uses Low-Rank Adaptation to slightly adapt the backbone’s internal representations instead of freezing it completely [52]. Applying the same strategy to the architecture proposed in this thesis is a natural next step, though it is worth noting that training even a small number of additional parameters, such as the roughly nine million trainable parameters used in that work, already exceeds what was comfortably achievable within the computational budget of this thesis, and most of the directions listed here would benefit from being pursued with fewer hardware constraints than the ones adopted throughout this work.

Two further directions follow directly from limitations identified earlier in this thesis rather than from external related work. Chapter 3 explicitly excluded fully regenerated generative models from the training data, since they provide no usable ground-truth mask; investigating whether a weaker form of supervision, such as an image-level authenticity label rather than a pixel-level mask, could still be exploited from this excluded category would substantially widen the pool of usable training data. Finally, the out-of-distribution evaluation of Chapter 5 is currently confined to a single unseen generative engine, Fooocus; extending this evaluation to a broader set of held-out commercial and open-source generators would provide a more robust estimate of real-world generalization.

Declaration of AI usage

The Guidelines for the Use of Artificial Intelligence of the University of Pavia mandate that any thesis for which AI tools have been used must include a declaration specifying the tools employed, the scope of application, and the manner of use.

For the preparation of this thesis, AI tools have been used for the following tasks:

- | | |
|-----------------------------------|--|
| Writing assistance: | Gemini 3.1 Pro, Claude Sonnet 5, and ChatGPT were employed to correct typographical errors, verify grammatical consistency, and suggest alternative formulations to improve readability and sentence variety. |
| Literature review: | Claude Sonnet 5 assisted in preliminary bibliographic research, particularly in Chapter 2, helping to identify and break down relevant references. Gemini 3.1 Pro was used to create a cohesive narrative about the historical progression of related works. ChatGPT was used for cross-verifying the correctness of text and references. |
| Structure and analysis: | Claude Sonnet 4.6, Claude Sonnet 5, and Gemini 3.1 Pro assisted in refining the outline and logical structure of selected chapters starting from the author's initial drafts. Additionally, they were used to verify the validity of the insights and analysis produced by the author, and to suggest additional research paths worth exploring. |
| Programming: | Claude Sonnet 4.6 and Gemini 3.1 Pro were used to assist in writing, debugging, and optimizing experimental scripts. They were also used to facilitate the process of learning functions of new libraries and GitHub usage. |
| Visual content generation: | Claude Sonnet 5 and Gemini 3.1 Pro were used to create the initial draft and layout of diagrams, tables, and illustrations. Additionally, they were employed for the production of the auxiliary scripts necessary for the creation of graphs and comparative images. |

In accordance with the University of Pavia Guidelines for the Use of Artificial Intelligence, the author declares that all AI-generated outputs used in the preparation of this thesis, including text, code, references, summaries, analyses, images, and other materials have been critically reviewed, verified, corrected where necessary, and integrated under the author's full responsibility. The final text and all academic claims remain the author's own work.

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